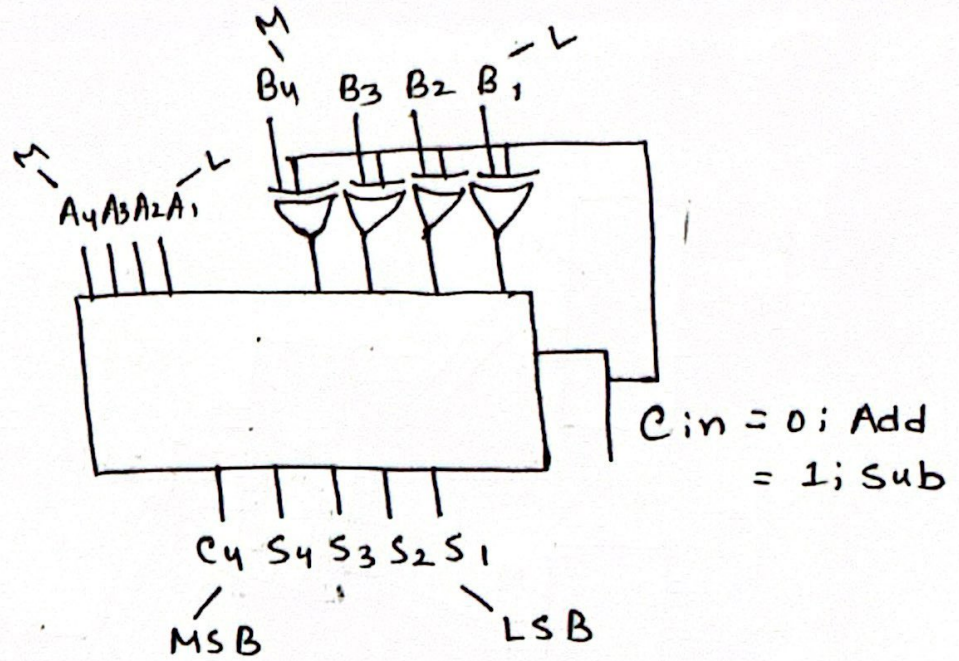


For all the following: Make sure that your circuit is efficient, meaning you should use the lowest number of components. You may use external gates if required.

- Build an adder cum subtractor (4 bits)
- Draw the block diagram of a 20 bits parallel adder.
- Build a 13 person voting system using full and parallel adders.
- Build a 8 person voting system using full and parallel adders.
- Build a 15 person voting system using full and parallel adders.
- Consider A is a 4 bit number. Design $A-3$ using a 4 bit parallel adder. Use external gates if required.
- Consider A is a 4 bit number. Design $A+3$ using a 4 bit parallel adder. Use external gates if required.
- Consider two numbers: 7 and 5. You can only calculate addition and subtraction between those two numbers. Design a circuit that can perform the above calculations based upon the user's intention.

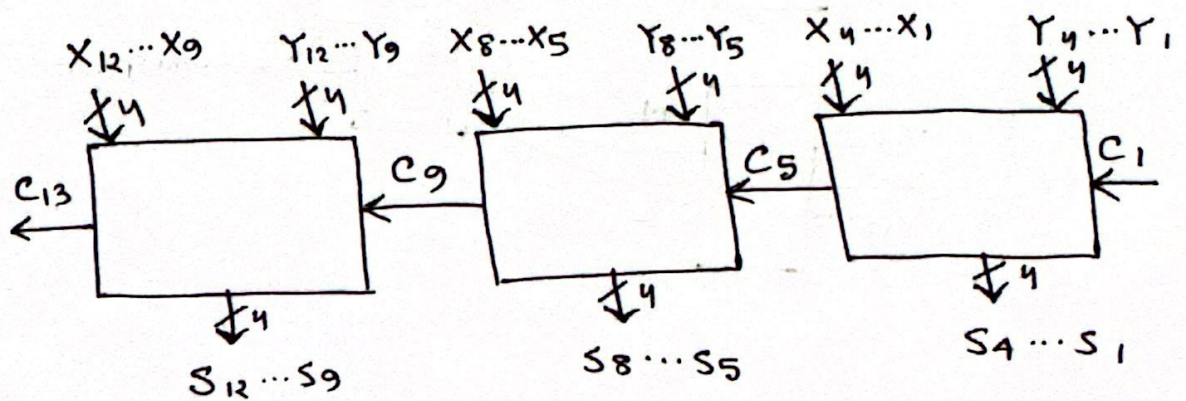
- Design a full adder using two half adders. You must use two NOR gates and no OR gates.
- Design a full adder using two half adders. You must use three NAND gates and no OR gates.
- Build a circuit that implements the 2's complement number system (3 bits) using encoder(s) and decoder(s).
- Design an Octal to Binary encoder.
- Design a 4x2 priority encoder prioritizing MSB.
- Design a 4x2 priority encoder prioritizing LSB.

①

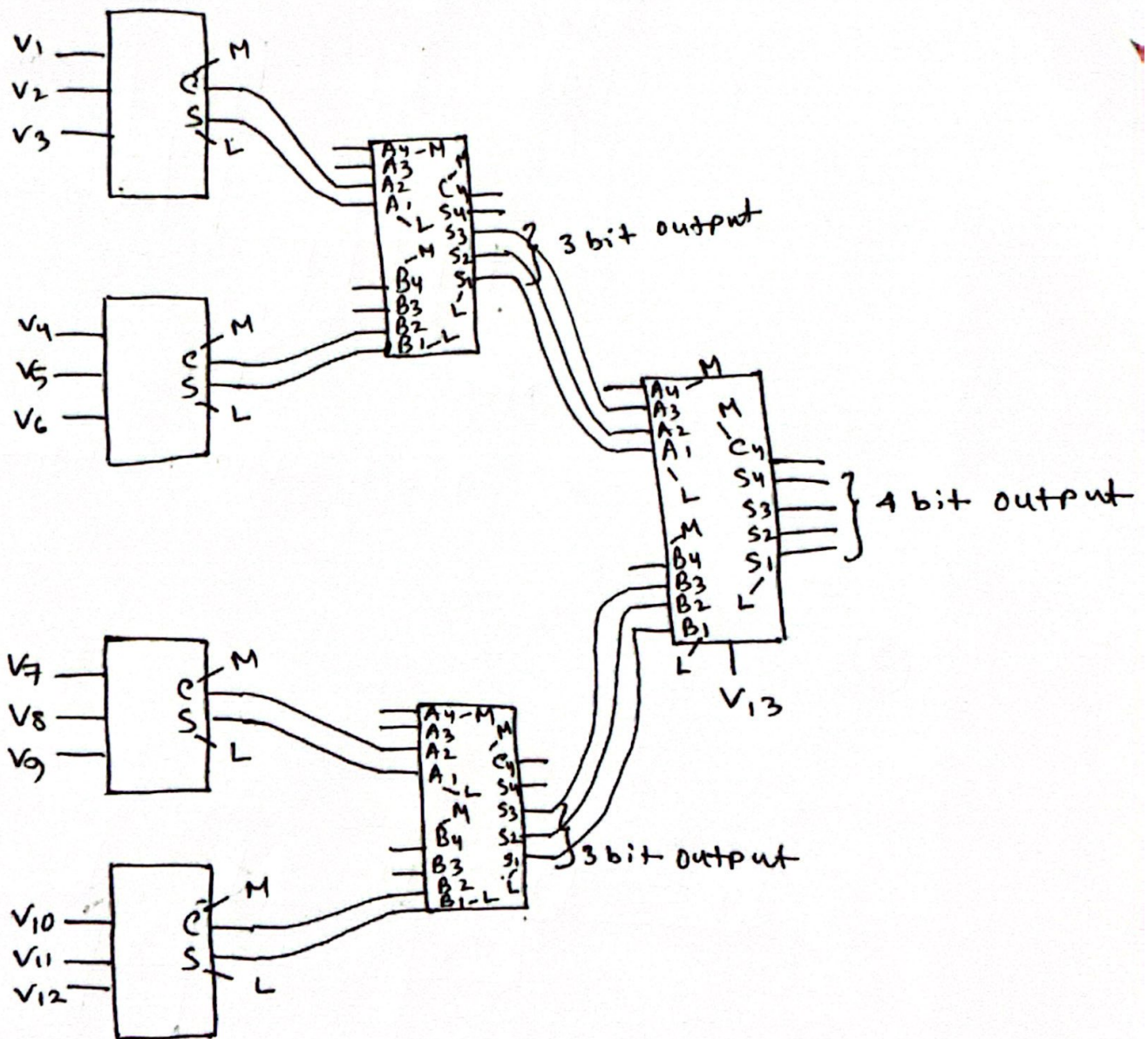


4 bit adder - subtractor

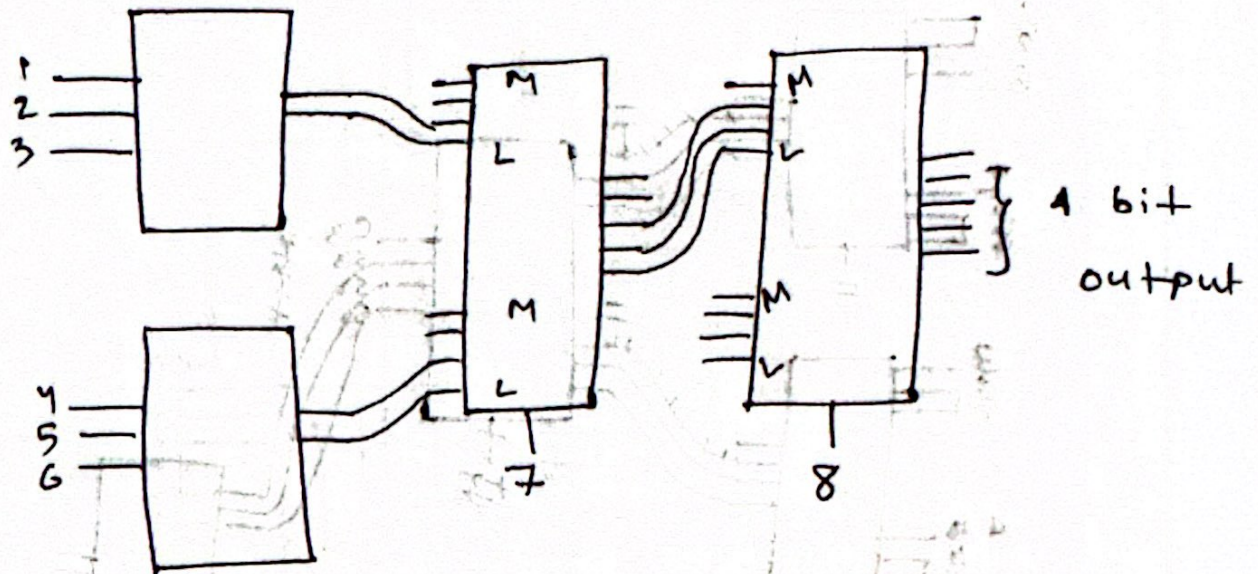
②



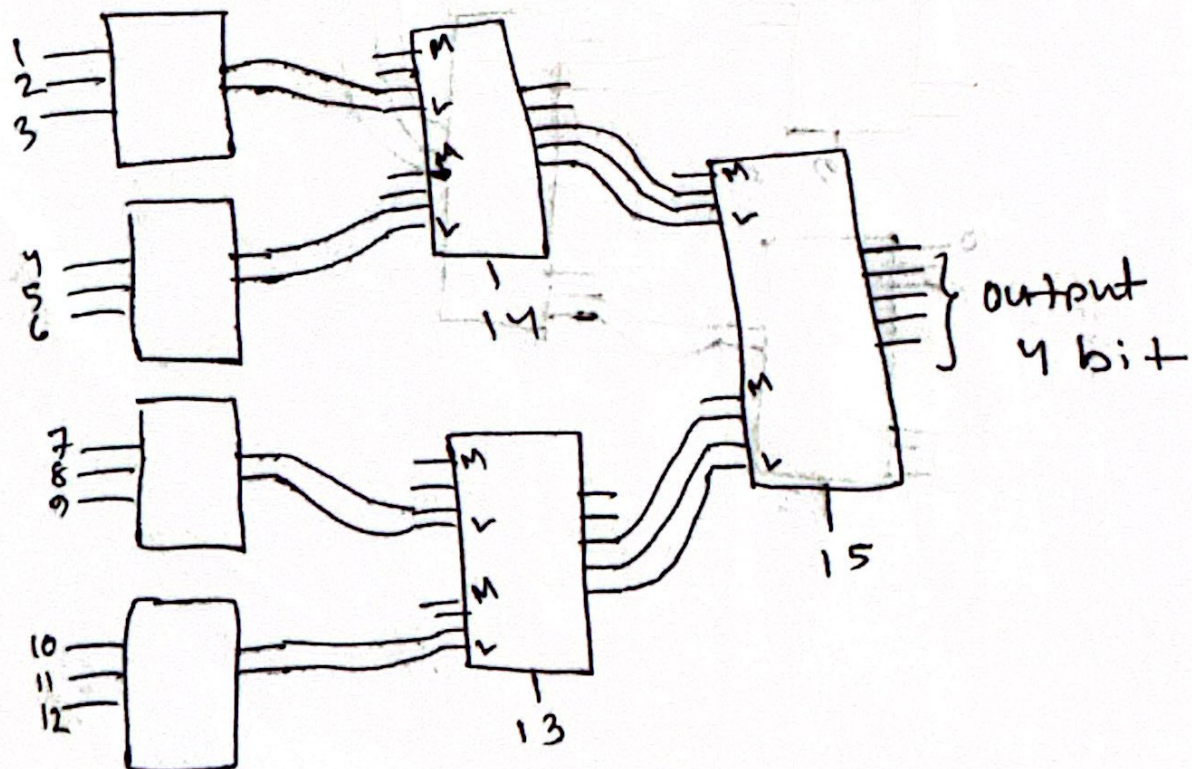
3



Build a 8 person voting system using full and parallel adder.



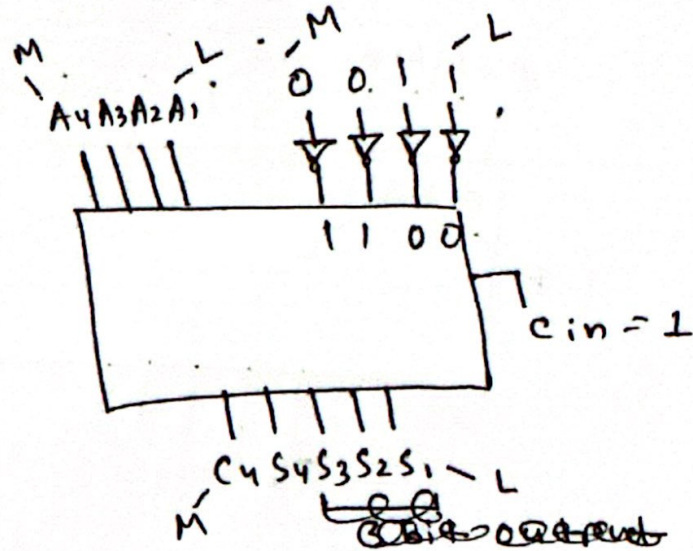
Build a 15 person voting system



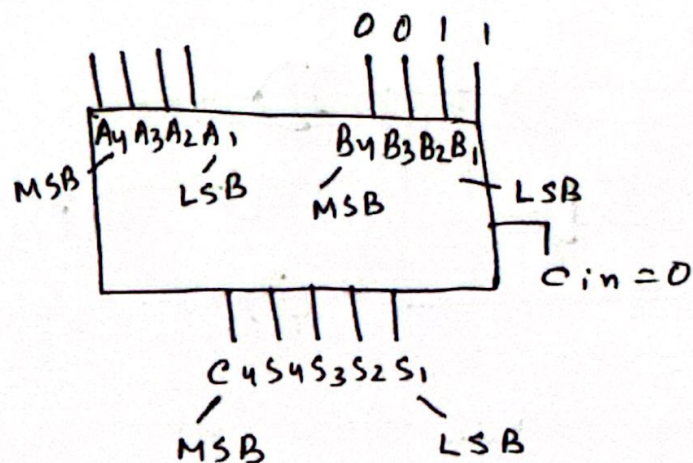
④ $A \rightarrow 4 \text{ bit}$

$$A - 3 = A + (-3)$$

$$\begin{array}{r} 3 \\ \downarrow \\ 0011 \\ \hline 1100 \\ +1 \\ \hline \end{array}$$



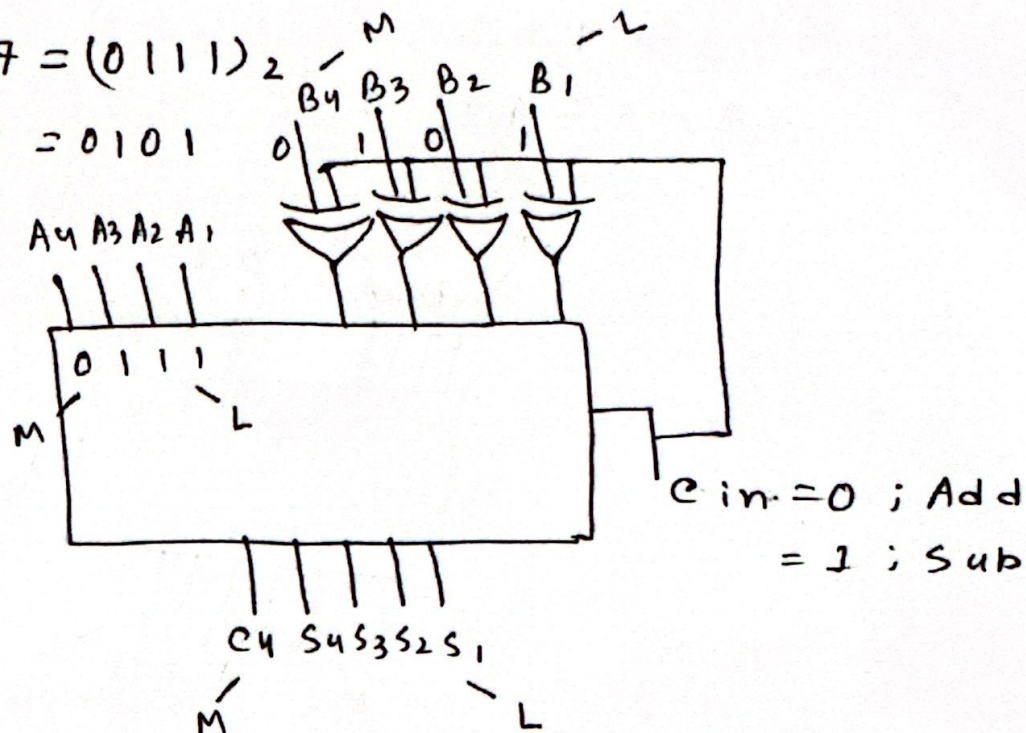
⑤



⑥

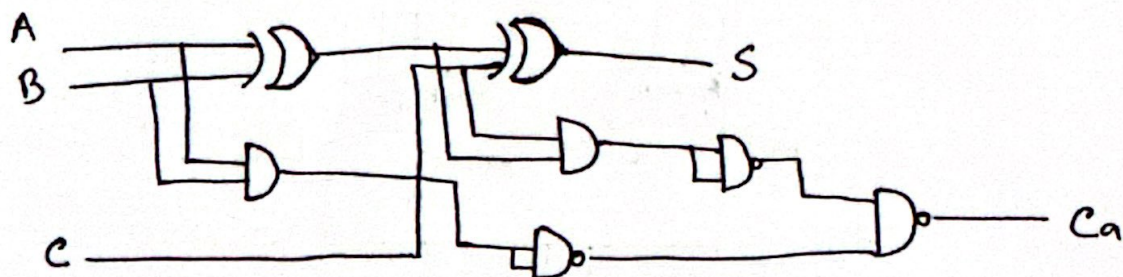
$$A = 7 = (0111)_2$$

$$B = 5 = 0101$$

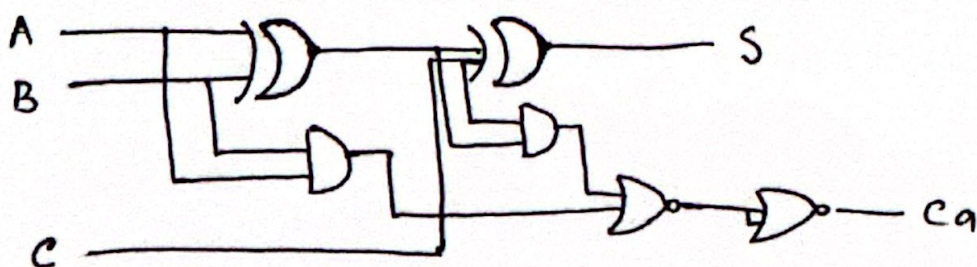


⑦

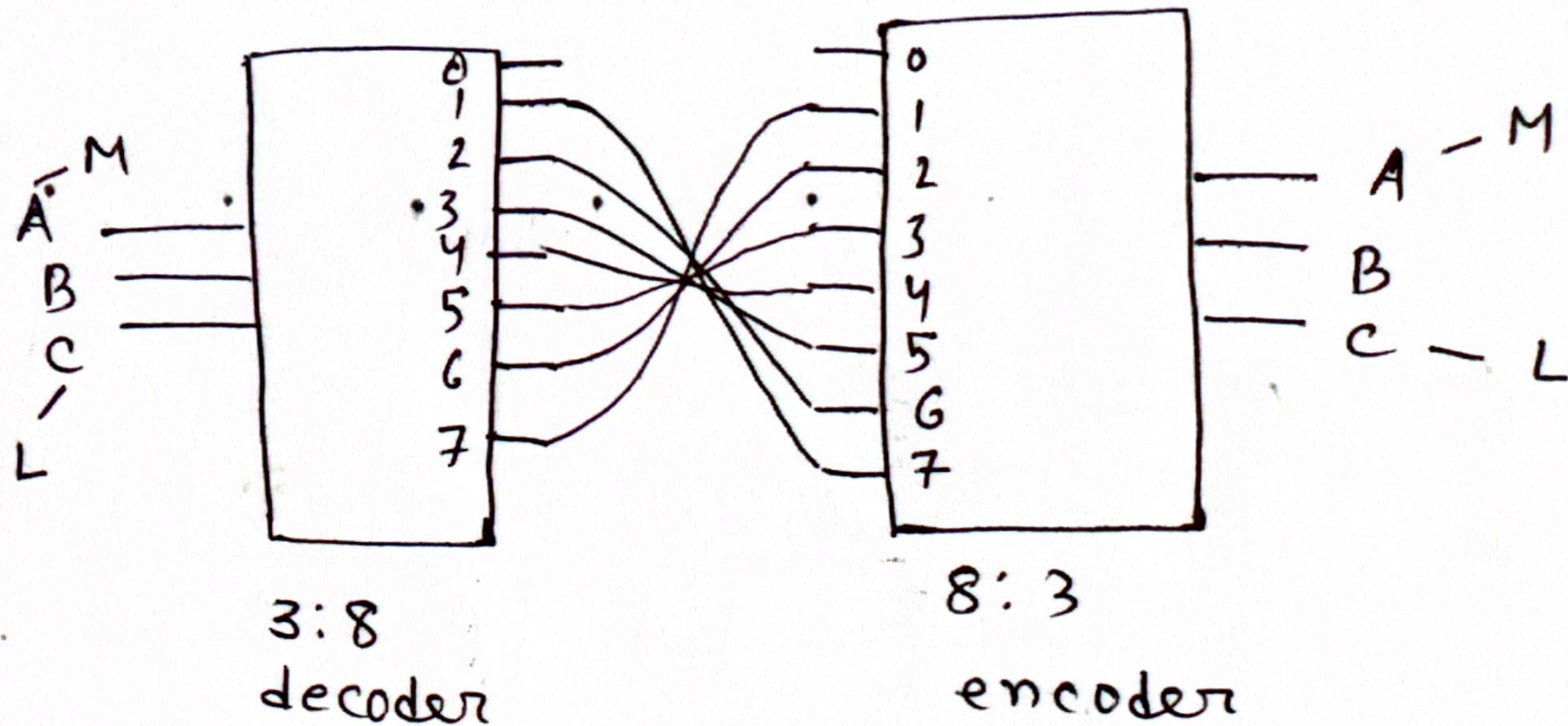
Using 3 NAND gate and no OR gates



Using 2 NOR gates and no OR gates



9



Example: Octal-binary encoder

TABLE 5-3

Truth Table of Octal-to-Binary Encoder

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$$z = D_1 + D_3 + D_5 + D_7$$

$$y = D_2 + D_3 + D_6 + D_7$$

$$x = D_4 + D_5 + D_6 + D_7$$

Example: Octal-binary encoder

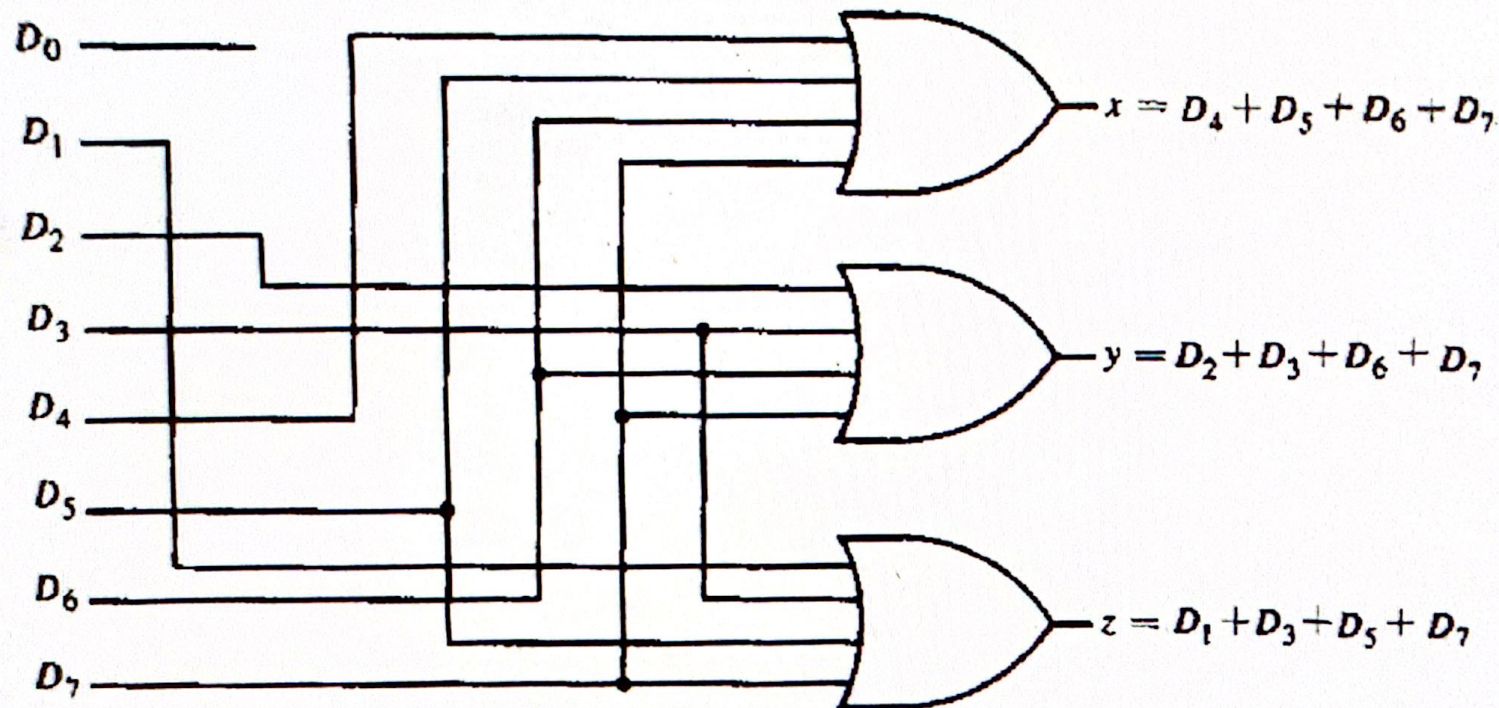


FIGURE 5-13

Octal-to-binary encoder

There are 8 input variables so there will 2^8 input combination, amongst which in Octal-binary encoder only 8 are useful

Design a 4X2 priority encoder prioritizing MSB.

I_3	I_2	I_1	I_0	A	B
0	0	0	0	x	x
0	0	0	1	0	0
0	0	1	0	0	1
0	0	1	1	0	1
0	1	0	0	1	0
0	1	0	1	1	0
0	1	1	0	1	0
0	1	1	1	1	0
1	0	0	0	1	1
1	0	0	1	1	1
1	0	1	0	1	1
1	0	1	1	1	1
1	1	0	0	1	1
1	1	0	1	1	1
1	1	1	0	1	1
1	1	1	1	1	1

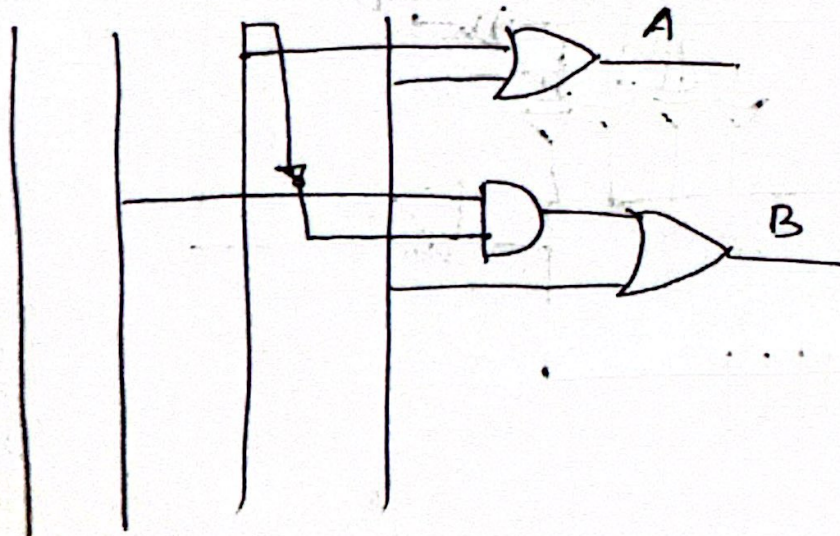
	$\bar{I}_1 \bar{I}_0$	$\bar{I}_1 I_0$	$I_1 \bar{I}_0$	$I_1 I_0$
$\bar{I}_3 \bar{I}_2$	0	1	3	2
$\bar{I}_3 I_2$	1	1	1	1
$I_3 \bar{I}_2$	1	1	1	1
$I_3 I_2$	1	1	1	1

$$A = I_2 + I_3$$

		1	1
1	1	1	1
1	1	1	1

$$B = I_3 + \bar{I}_1 \bar{I}_2$$

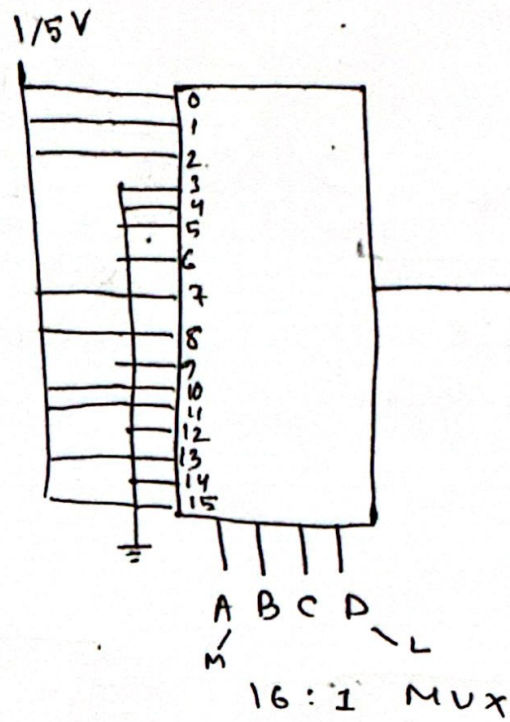
I_0 I_1 I_2 I_3



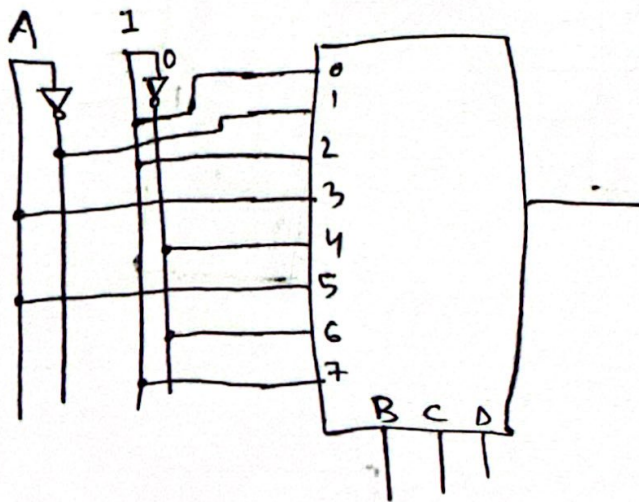
- Implement the following boolean function using a single 16:1 mux .
 $F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$. Use external gates if required.
- Implement the following boolean function using a single 8:1 mux.
 $F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$. Use external gates if required.
- Implement the following boolean function using a single 4:1 mux.
 $F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$. Use external gates if required.
- Implement the following boolean function using both 4:1 and 2:1 mux in a single circuit
 $F(A,B,C,D) = \sum(0,1,2,7,8,10,11,13, 15)$. Use external gates if required.
- Implement the following boolean function using a) 4x16 decoder(s) only b) 2x4 decoder(s) only
 $F(A,B,C,D,E) = \sum(0,1,2,7,8,10,11,13, 15,18,21,24,25)$. Use external gates if required.
- Build a full adder using encoder(s) and decoder(s).
- Build a BCD to Excess-3 code converter using encoder(s) and decoder(s).
- Build a BCD to Excess-5 code converter using encoder(s) and decoder(s).
- Design the circuit diagram for a 3x8 decoder.
- Design the circuit diagram for a 2x4 decoder.
- Design the circuit diagram for a 4:1 mux.
- Design a 8:1 mux using both 4:1 mux and 2:1 mux in a single circuit.
- Design AND, OR, NOT gate using 4:1 mux.
- Design AND, OR, NOT gate using 2:1 mux.
- Design a circuit that can demonstrate the usage of ENABLE pin in decoders.

⑩

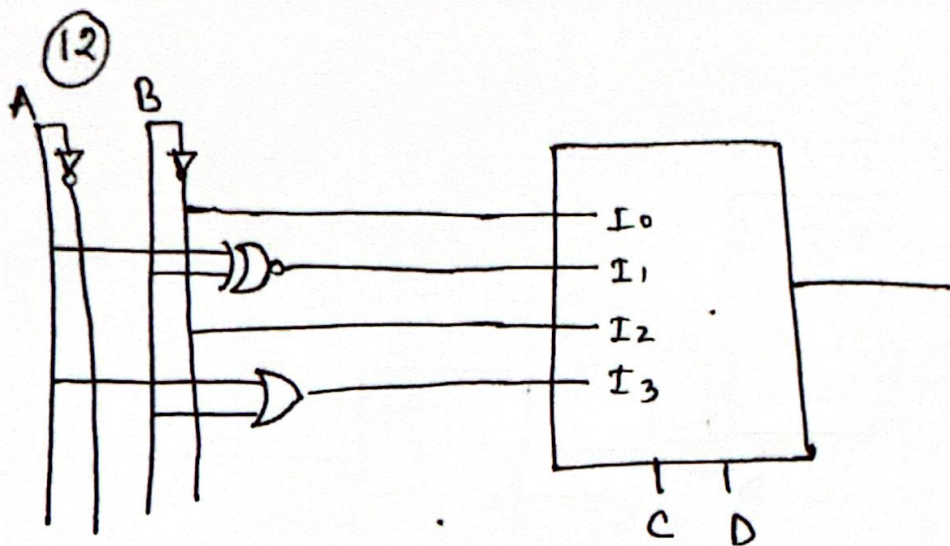
$$F(A, B, C, D) = \sum(0, 1, 2, 7, 8, 10, 11, 13, 15)$$



⑪



	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	1	A'	1	A	0	A	0	1



	I ₀	I ₁	I ₂	I ₃
A'B'	0	1	2	3
A'B	4	5	6	7
AB'	8	9	10	11
AB	12	13	14	15
	B'	A ⊙ B	B'	A + B

I₀ ⇒

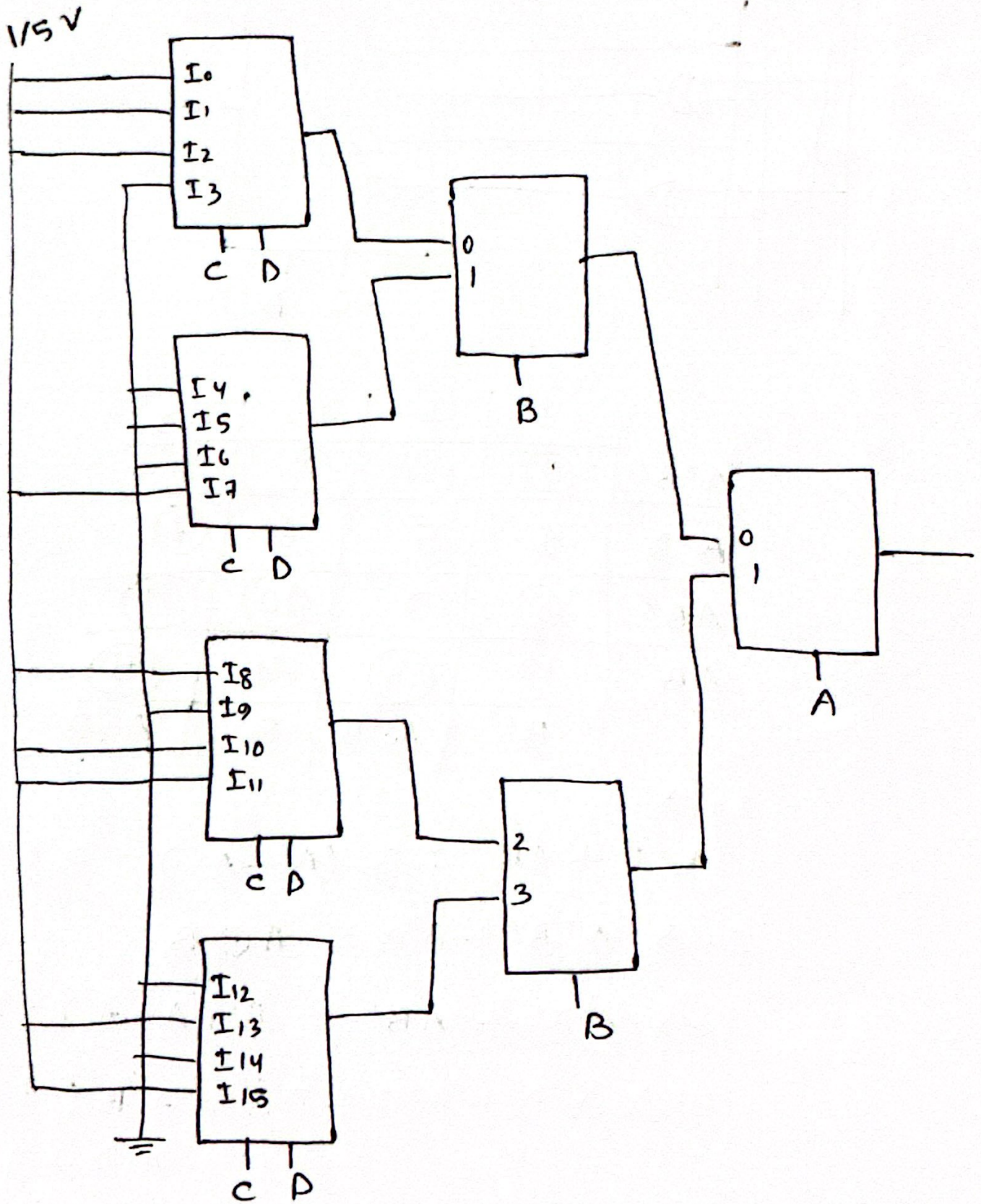
$$A'B' + AB' = B'(A + A') = B'$$

I₁ ⇒ $A'B' + AB = A ⊙ B$

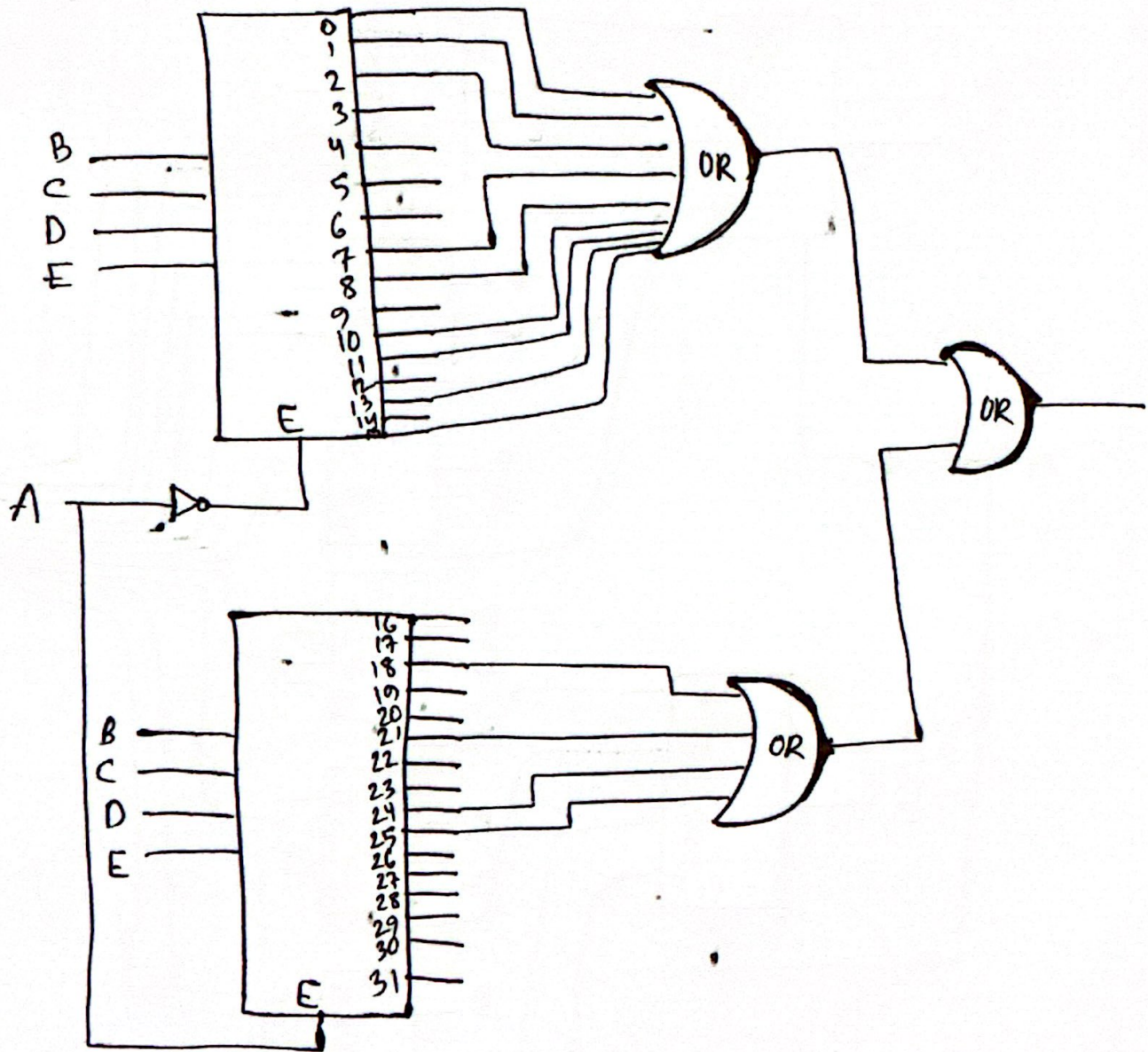
I₂ ⇒ $A'B' + AB' = B'(A + A') = B'$

$$\begin{aligned}
 I_3 \Rightarrow A'B + AB' + AB &= A'B + A(B + B') \\
 &= A'B + A \\
 &= (A' + A)(B + A) \\
 &= A + B
 \end{aligned}$$

13

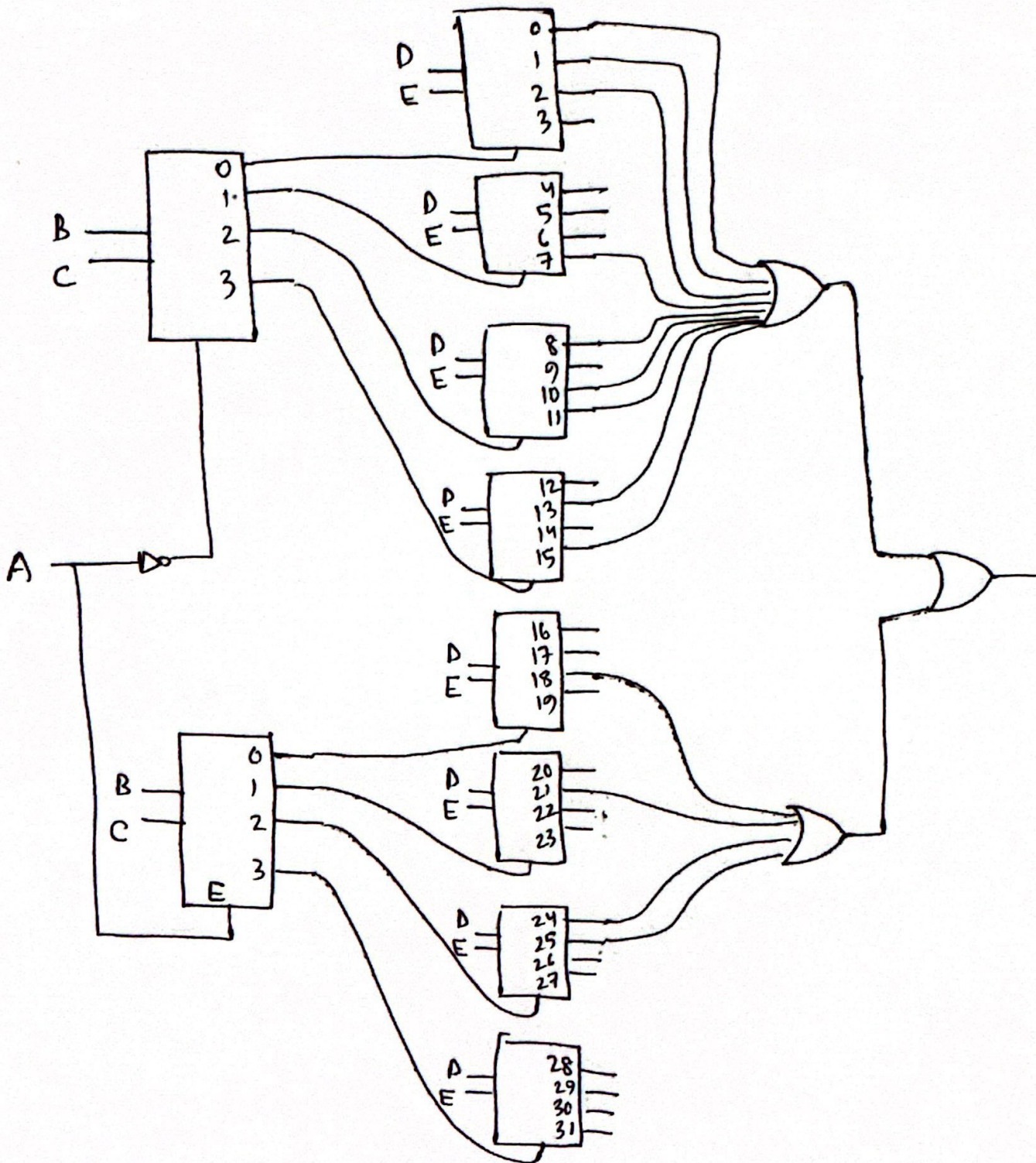


(14) a)

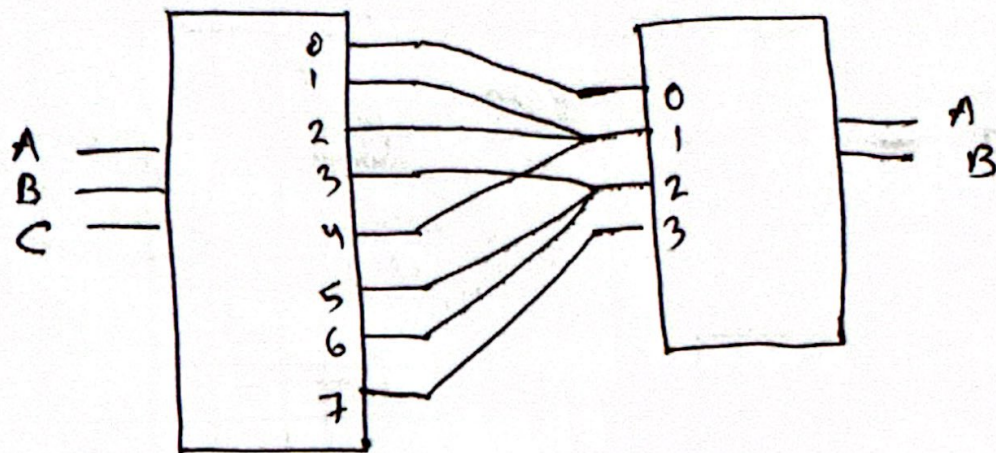


(A)

b)

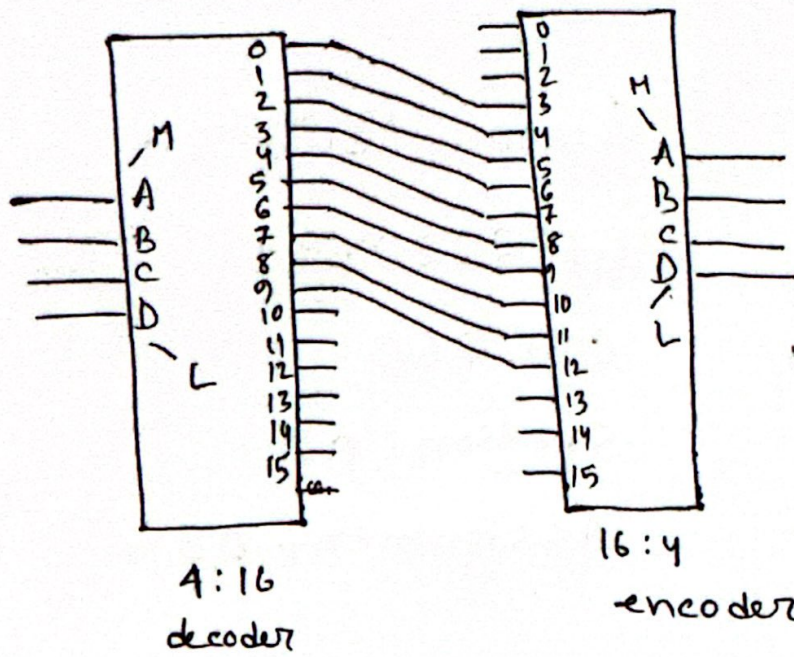


7) Design a full adder using encoder and decoder.

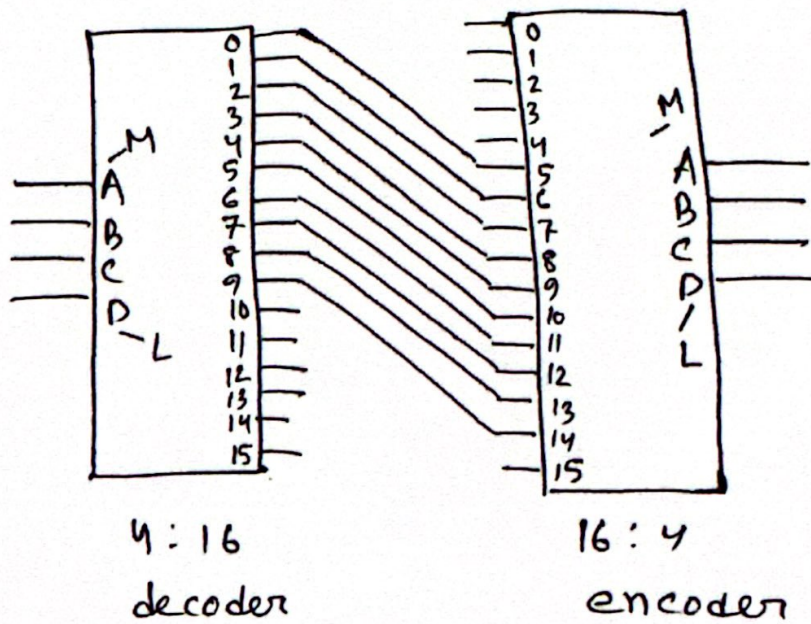


A	B	C	Ca	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

15



16



* Design the circuit diagram for a

3x8 decoder.

1010 = 2 = 2

	$\overline{A}$	B	$\overline{C}$	O_0	O_1	O_2	O_3	O_4	O_5	O_6	O_7
0	0	0	0	1	0	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0
2	0	1	0	0	0	1	0	0	0	0	0
3	0	1	1	0	0	0	1	0	0	0	0
4	1	0	0	0	0	0	0	1	0	0	0
5	1	0	1	0	0	0	0	0	1	0	0
6	1	1	0	0	0	0	0	0	0	1	0
7	1	1	1	0	0	0	0	0	0	0	1

$$O_0 = A'B'C'$$

$$O_1 = A'B'C$$

$$O_2 = A'BC'$$

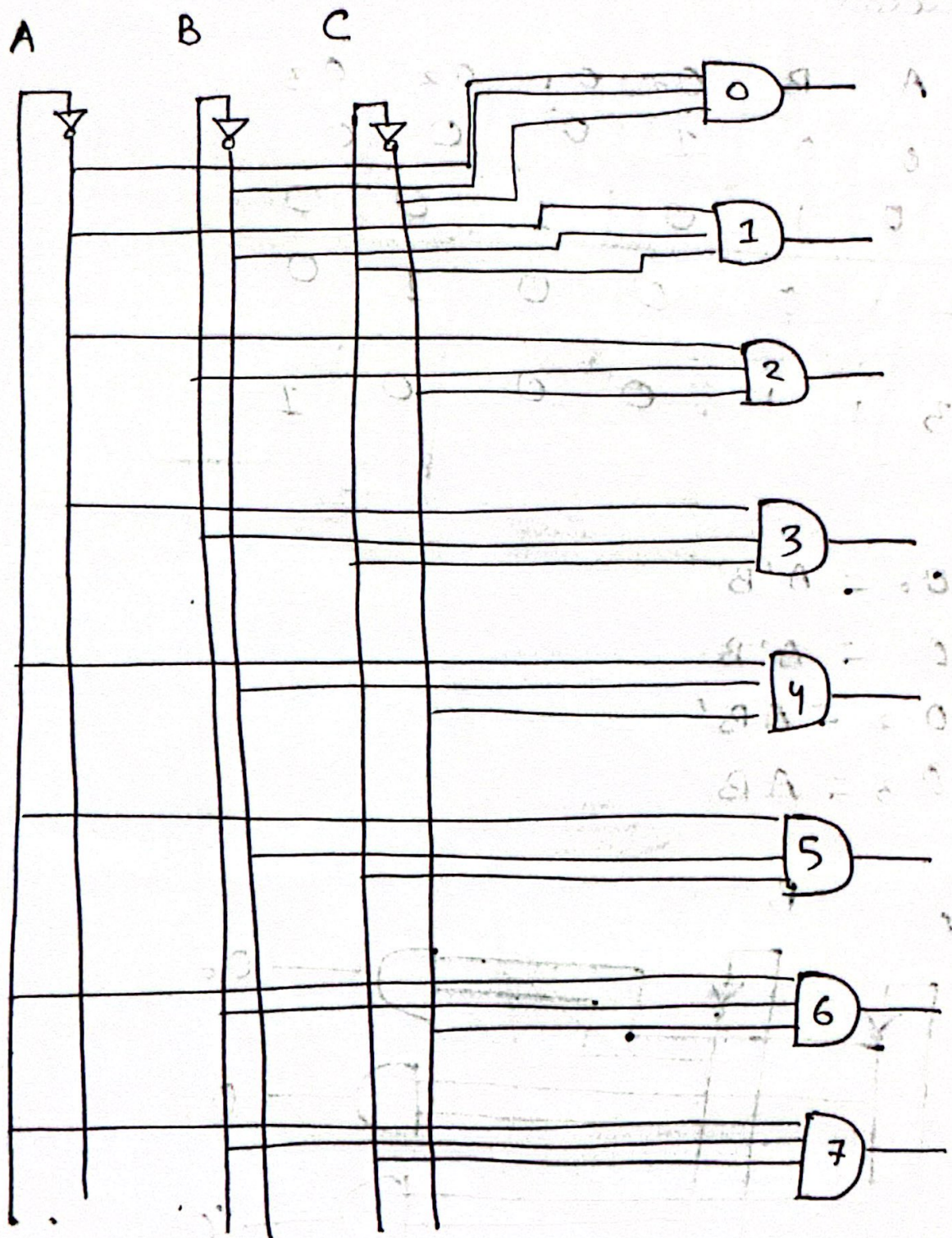
$$O_3 = A'BC$$

$$O_4 = AB'C'$$

$$O_5 = AB'C$$

$$O_6 = ABC'$$

$$O_7 = ABC$$



Design the circuit diagram for a 2X4 decoder.

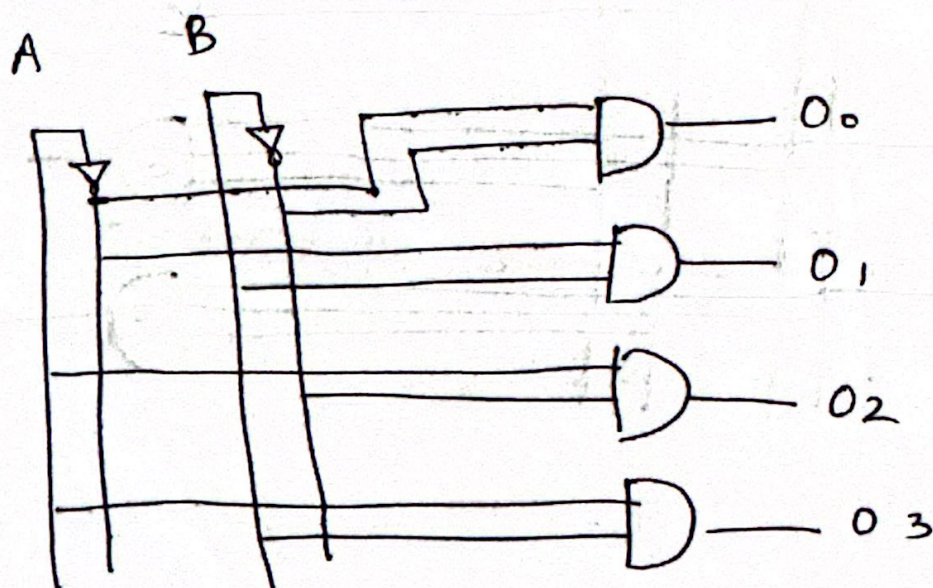
	A	B	O_0	O_1	O_2	O_3
0	0	0	1	0	0	0
1	0	1	0	1	0	0
2	1	0	0	0	1	0
3	1	1	0	0	0	1

$$O_0 = A'B'$$

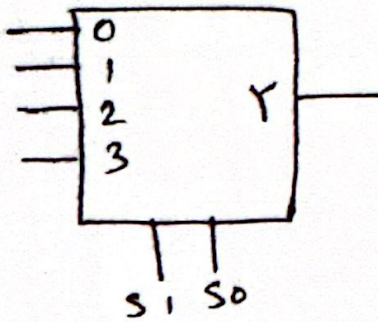
$$O_1 = A'B$$

$$O_2 = AB'$$

$$O_3 = AB$$

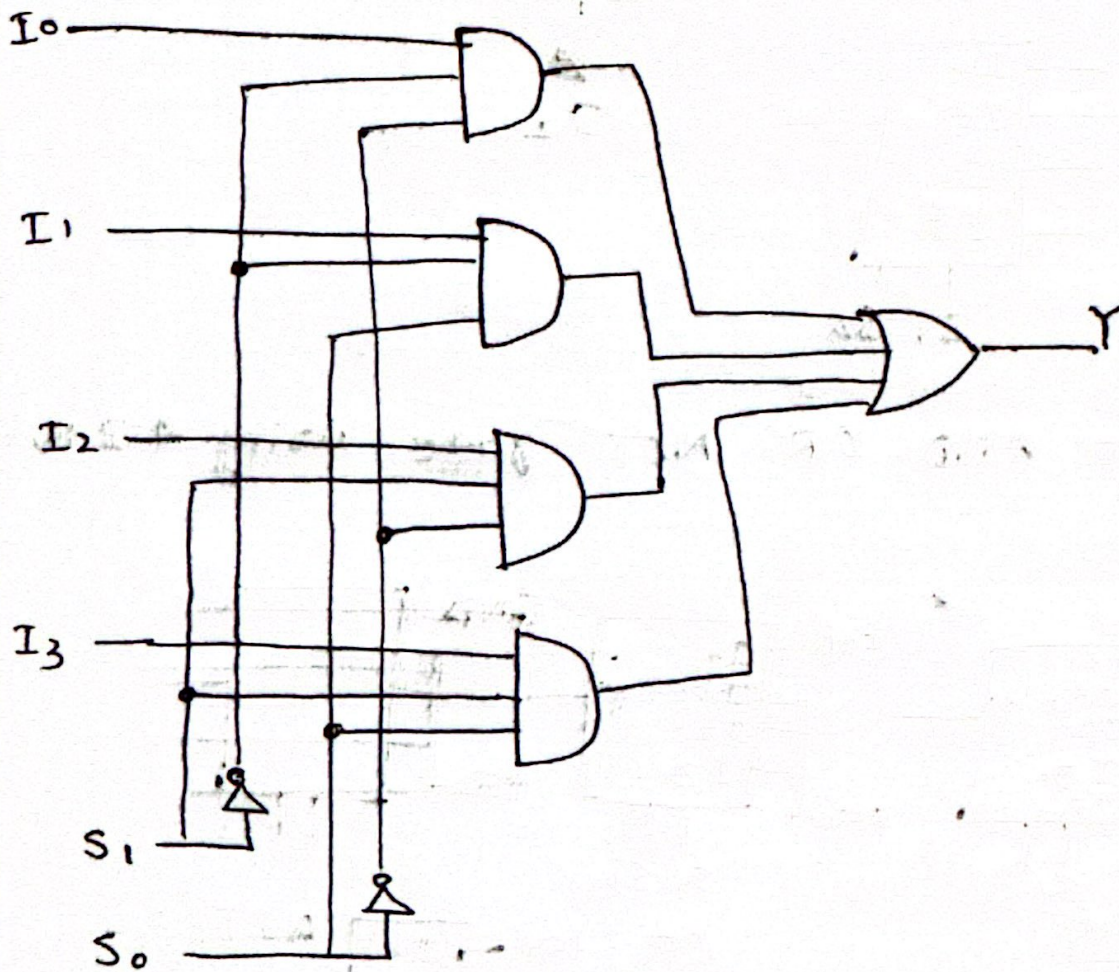


Design the circuit diagram for a 4:1 mux

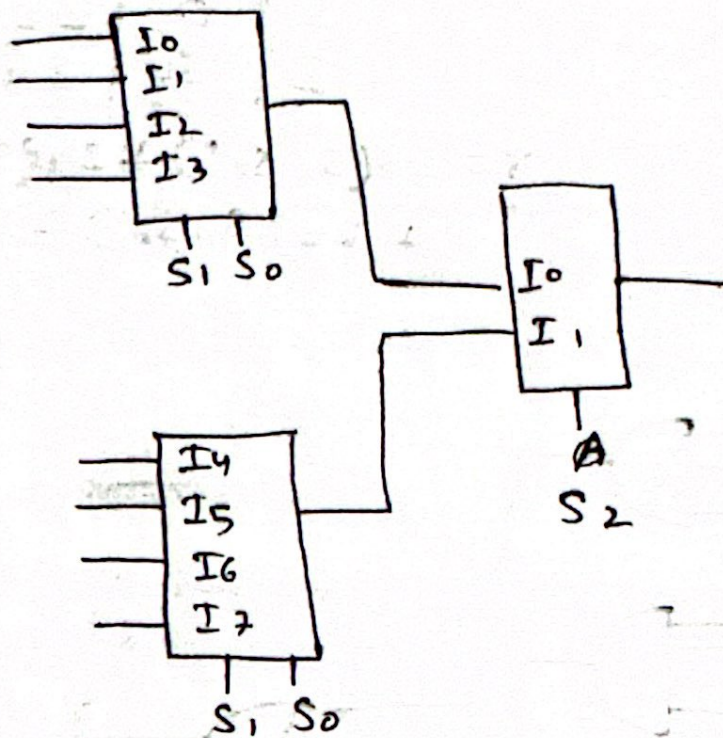


S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

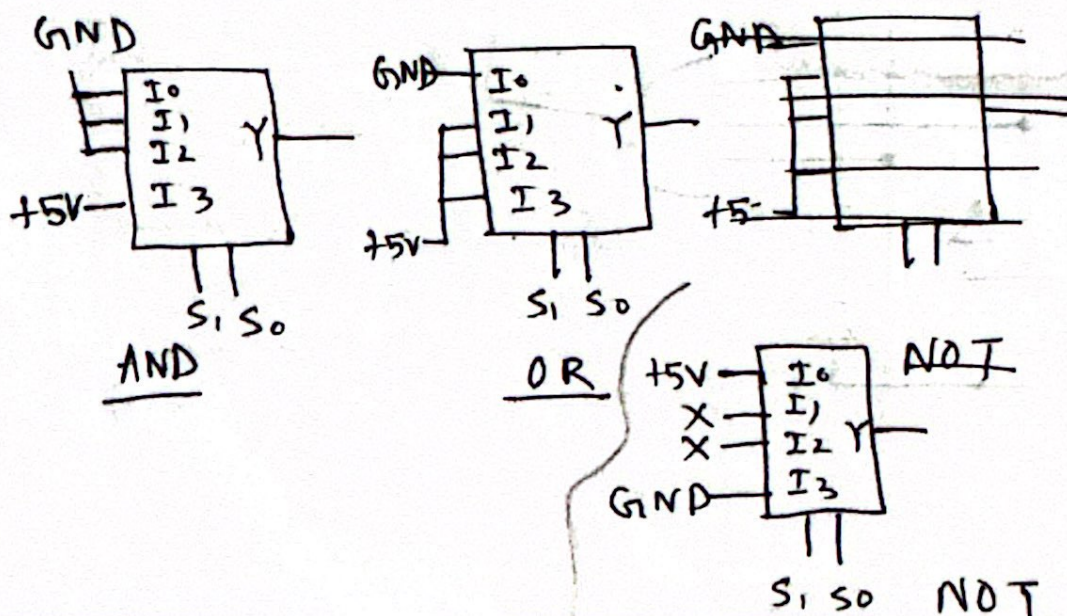
$$Y = I_0(S_1'S_0') + I_1(S_1'S_0) + I_2(S_1S_0') + I_3(S_1S_0)$$



Design 8:1 mux using both 4:1 mux and 2:1 mux in a single circuit.

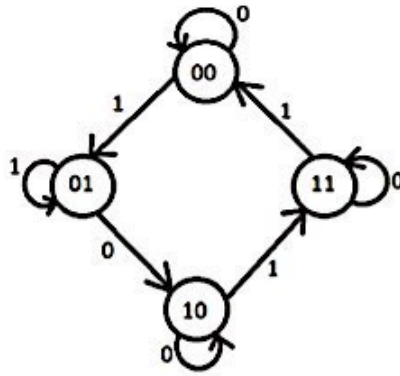


Design AND, OR, NOT gate using 4:1 mux

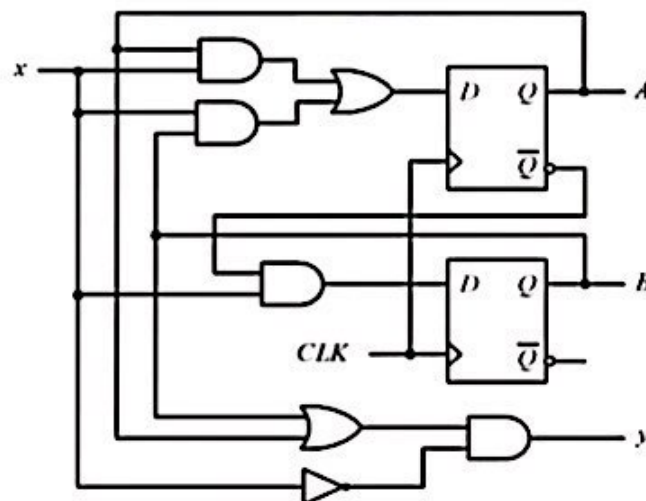


- Design a D FF using SR FF.
- Design a T FF using JK FF.
- Design a D FF using JK FF.
- Design a SR FF.
- Write down the excitation table for SR, D, JK and T FF.
- Write down the characteristics table for SR, D, JK and T FF.

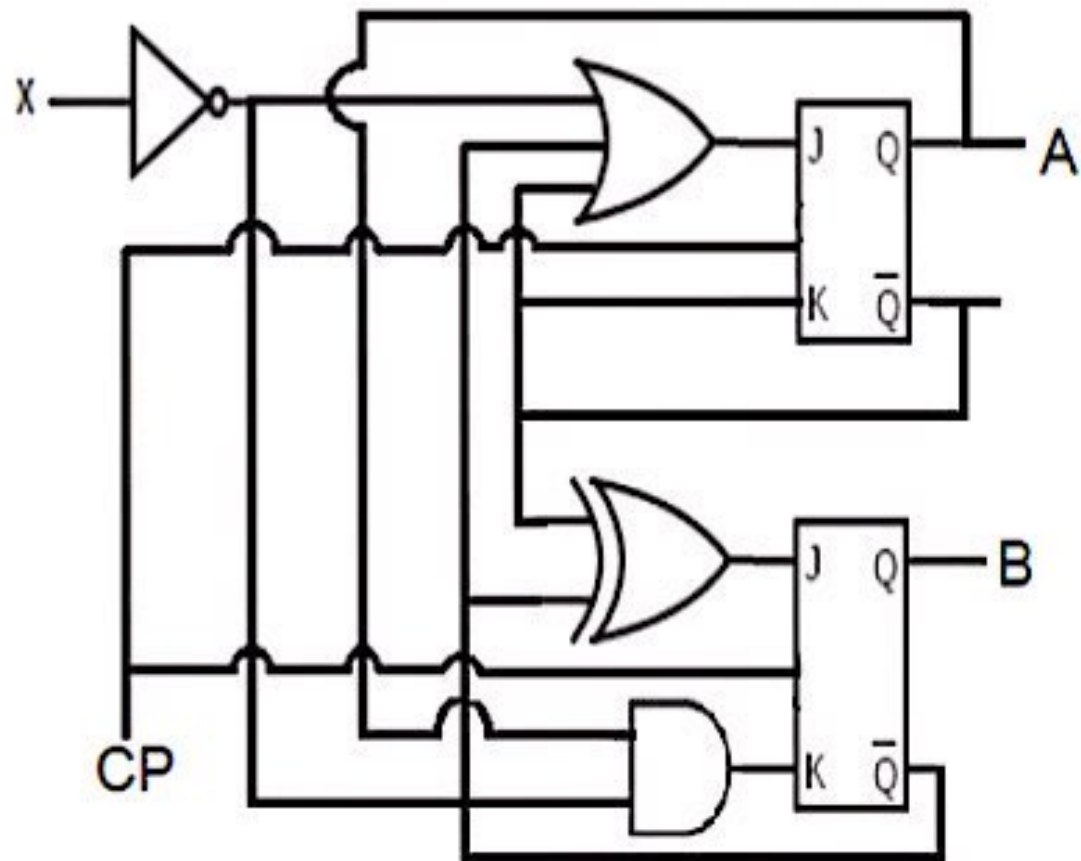
- Given the state diagram as follows, get the sequential circuit using SR flipflop.



- Implement the following counter using T flip flop
CSE110 -> CSE111 -> CSE220 -> CSE221 -> CSE331 -> CSE221 -> CSE321 -> CSE110
 - 3->4->6->10->12->13->15->3
 - Implement the given counter using JK flip-flop.
 - Implement the given counter using T flip-flop.
- NB: For states not given in question, please move to the initial state as per question.
- Implement 4 bit up counter using JK flip flop
 - Implement 4 bit down counter using JK flip flop
 - Implement 2 bit up/down counter using D flip flop
 - Implement 3 bit up/down counter using T flip flop
 - Implement 2 bit up/down counter using SR flip flop
 - Implement 3 bit up/down counter using JK flip flop
 - Implement the following counter using T FF:
Green->Orange->Yellow->Red->Yellow->Orange->Yellow->Green
 - Implement the following counter using JK FF: Green->Yellow->Red->Yellow->Green
 - Implement the following counter using JK FF: 1->2->3->5->7->11->13->1
 - Implement the following counter using SR FF: 2->3->6->8->10->12->2
 - Draw the state diagram for the given circuit.



- Draw the state diagram for the given circuit.

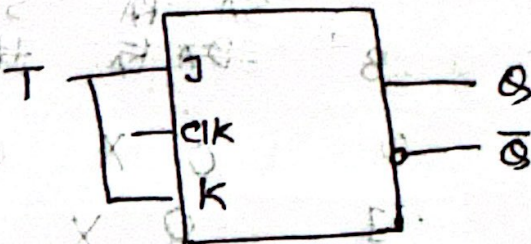


Assignment 4

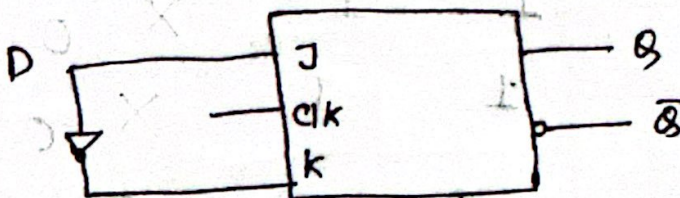
1. Design a D FF using SR FF



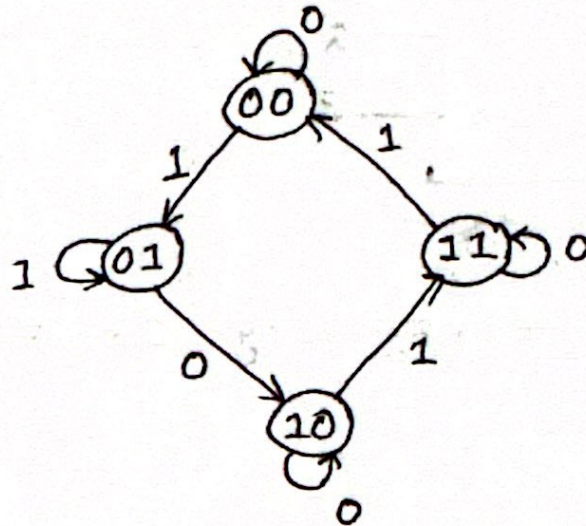
2. Design a T FF using JK FF.



3. Design a D IFF using JK FF



④ Given the state diagram as follows, get the sequential circuit using SR flipflop.



A	B	α	A^+	B^+	SA	RA	SB	RB
0	0	0	0	0	0	X	0	X
0	0	1	0	1	0	X	1	0
0	1	0	1	0	1	0	0	1
0	1	1	0	1	0	X	X	0
1	0	0	1	0	X	0	0	X
1	0	1	1	1	X	0	1	0
1	1	0	1	1	X	0	X	0
1	1	1	0	0	0	1	0	1

SR flip flop excitation table

$Q(t)$	$Q(t+1)$	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

$$S_A = \Sigma(2, 4, 5, 6)$$

$$R_A = \Sigma(0, 1, 3, 7)$$

$$S_B = \Sigma(1, 3, 5, 6)$$

$$R_B = \Sigma(1, 3, 5, 8)$$

	$\bar{B}\bar{X}$	$\bar{B}X$	BX	$B\bar{X}$
$\bar{A}$	0	1	3	2
A	4	5	7	6

$$S_A = B\bar{X}$$

	$\bar{B}\bar{X}$	$\bar{B}X$	BX	$B\bar{X}$
$\bar{A}$	X	X	X	
A			1	

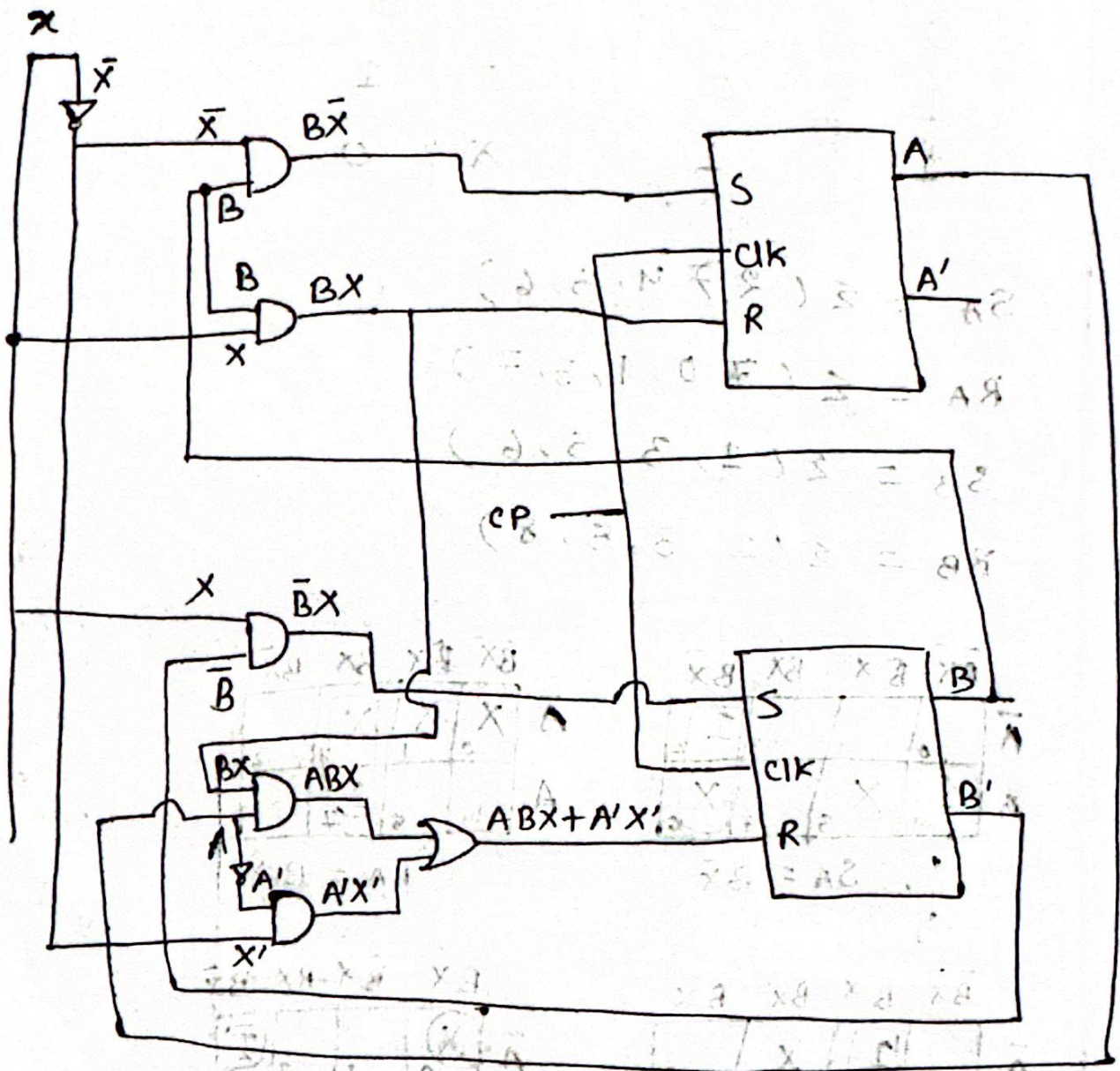
$$R_A = B\bar{X}$$

	$\bar{B}\bar{X}$	$\bar{B}X$	BX	$B\bar{X}$
$\bar{A}$	0	1	3	2
A	4	5	7	6

$$S_B = \bar{B}X$$

	$\bar{B}\bar{X}$	$\bar{B}X$	BX	$B\bar{X}$
$\bar{A}$	X			1
A	X		1	

$$R_B = ABX + \bar{A}B\bar{X}$$



⑧

CSE 110 $\rightarrow$ 000

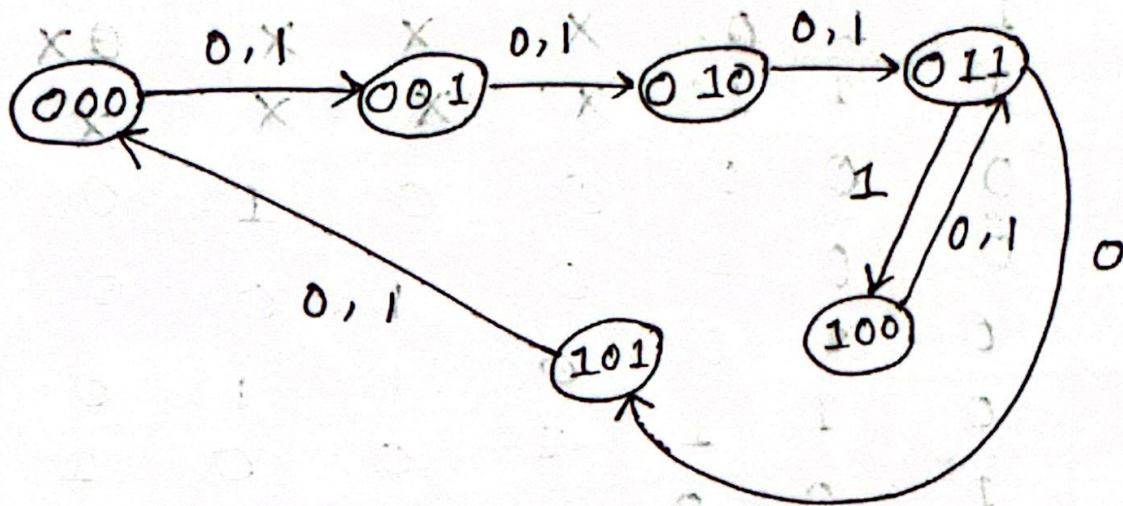
CSE 111 $\rightarrow$ 001

CSE 220 $\rightarrow$ 010

CSE 221 $\rightarrow$ 011

CSE 331 $\rightarrow$ 100

CSE 321 $\rightarrow$ 101



x	A	B	C	A^+	B^+	C^+	T_A	T_B	T_C
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	1	1	1	0
0	1	0	0	0	0	0	1	1	1
0	1	0	1	0	0	0	1	0	1
0	1	1	0	X	X	X	X	X	X
0	1	1	1	X	X	X	X	X	X
1	0	0	0	0	0	1	0	0	1
1	0	0	1	0	1	0	0	1	1
1	0	1	0	0	1	1	0	0	1
1	0	1	1	1	0	0	1	1	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	0	0	0	1	0	1
1	1	1	0	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X

$$T_A = \{3, 4, 5, 11, 12, 13\}$$

$$T_B = \{1, 3, 4, 9, 11, 12\}$$

$$T_C = \{0, 1, 2, 4, 5, 8, 9, 10, 11, 12, 13\}$$

ΣA	BC	$B'C'$	$B'C$	BC	BC'			
$\Sigma A'$	0	1	1	3	2			
ΣA	1	4	1	5	X	X	6	
ΣA	1	12	1	13	X	15	X	14
$\Sigma A'$	8	9	1	11	10			

$$T_A = A + BC$$

ΣA	BC	$B'C'$	$B'C$	BC	BC'			
$\Sigma A'$	0	1	1	3	2			
ΣA	1	4	1	5	X	X	6	
ΣA	1	12	1	13	X	15	X	14
$\Sigma A'$	8	9	1	11	10			

$$T_B = A'C + \cancel{BC}$$

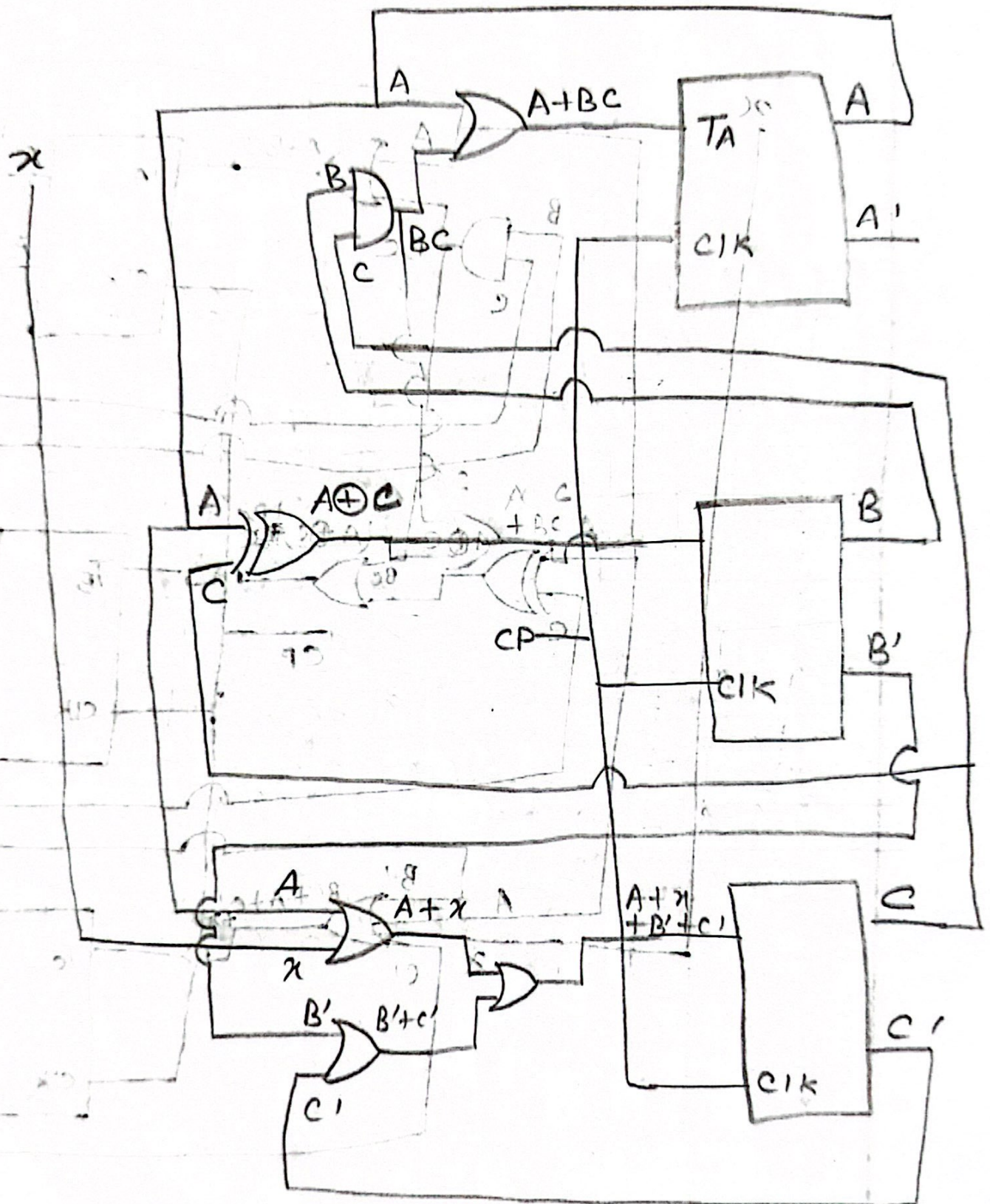
$$+ AC'$$

$$= (A \oplus C) + \cancel{BC}$$

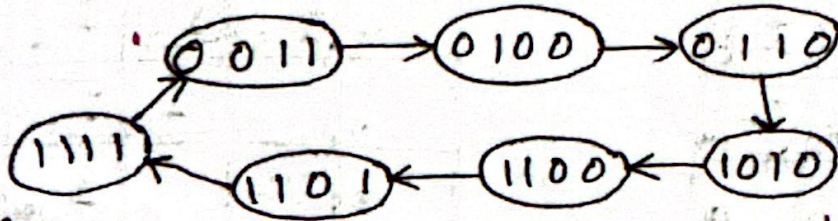
ΣA	BC	$B'C'$	$B'C$	BC	BC'			
$\Sigma A'$	1	1	3	1	2			
ΣA	1	4	1	5	X	X	6	
ΣA	1	12	1	13	X	15	X	14
$\Sigma A'$	1	1	1	1	1	10		

$$T_C = B' + A$$

$$+ C' + \cancel{A}$$



⑨. $3 \rightarrow 4 \rightarrow 6 \rightarrow 10 \rightarrow 12 \rightarrow 13 \rightarrow 15 \rightarrow 3$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $0011 \quad 0100 \quad 0110 \quad 1010 \quad 1100 \quad 1101 \quad 1111 \quad 0011$



A_3	A_2	A_1	A_0	A_3^+	A_2^+	A_1^+	A_0^+	TA_3	TA_2	TA_1	TA_0
0	0	0	0	0	0	1	1	0	0	1	1
0	0	0	1	0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	1	0	0	0	1	0
0	1	0	1	0	0	1	1	0	1	1	0
0	1	1	0	1	0	1	0	1	1	0	0
0	1	1	1	0	0	1	1	0	1	0	0
1	0	0	0	0	0	1	1	1	0	1	1
1	0	0	1	0	0	1	1	1	0	1	0
1	0	1	0	1	1	0	0	0	1	1	0
1	0	1	1	0	0	1	1	1	0	0	0
1	1	0	0	1	1	0	1	0	0	0	1
1	1	0	1	1	1	1	1	0	0	1	0
1	1	1	0	0	0	1	1	1	1	0	1
1	1	1	1	0	0	1	1	1	1	0	0

		$A_1 A_0$			
		$\bar{A}_1 \bar{A}_0$	$\bar{A}_1 A_0$	$A_1 \bar{A}_0$	$A_1 A_0$
$A_3 A_2$	$\bar{A}_3 \bar{A}_2$	0	1	3	2
	$\bar{A}_3 A_2$	4	5	7	6
	$A_3 \bar{A}_2$	8	9	11	10
	$A_3 A_2$	12	13	15	14

$$T_A = A_3 \bar{A}_2 + A_1 A_0 + A_1 \bar{A}_0$$

		$A_1 A_0$			
		$\bar{A}_1 \bar{A}_0$	$\bar{A}_1 A_0$	$A_1 \bar{A}_0$	$A_1 A_0$
$A_3 A_2$	$\bar{A}_3 \bar{A}_2$	1	0	1	3
	$\bar{A}_3 A_2$	1	4	1	5
	$A_3 \bar{A}_2$	1	8	1	9
	$A_3 A_2$	1	12	1	13

$$T_{A_1} = \bar{A}_3 \bar{A}_1 + \bar{A}_1 A_0 + A_3 \bar{A}_2 \bar{A}_0 + \bar{A}_3 \bar{A}_2 A_0$$

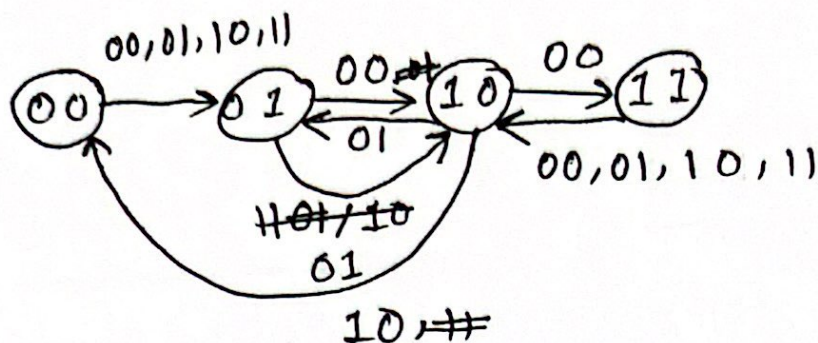
		$A_1 A_0$			
		$\bar{A}_1 \bar{A}_0$	$\bar{A}_1 A_0$	$A_1 \bar{A}_0$	$A_1 A_0$
$A_3 A_2$	$\bar{A}_3 \bar{A}_2$	1	0	1	3
	$\bar{A}_3 A_2$	1	4	1	5
	$A_3 \bar{A}_2$	1	8	1	9
	$A_3 A_2$	1	12	1	13

$$T_{A_2} = A_1 A_2 + A_3 \bar{A}_1 \bar{A}_0 + \bar{A}_3 A_2 A_0 + A_1 A_0 \bar{A}_3$$

		$A_1 A_0$			
		$\bar{A}_1 \bar{A}_0$	$\bar{A}_1 A_0$	$A_1 \bar{A}_0$	$A_1 A_0$
$A_3 A_2$	$\bar{A}_3 \bar{A}_2$	1	0	1	3
	$\bar{A}_3 A_2$	1	4	1	5
	$A_3 \bar{A}_2$	1	8	1	9
	$A_3 A_2$	1	12	1	13

$$T_{A_0} = A_3 \bar{A}_1 \bar{A}_0 + \bar{A}_3 \bar{A}_2 A_1 + \bar{A}_3 \bar{A}_2 \bar{A}_0 + A_3 A_2 \bar{A}_0$$

(12) Green $\rightarrow$ Orange $\rightarrow$ yellow $\rightarrow$ red $\rightarrow$ yellow $\rightarrow$ orange
 $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$ $\rightarrow$ yellow $\rightarrow$ green
 00 01 10 11



x	y	A	B	A+	B+	T _A	T _B
0	0	0	0	0	1	0	1
0	0	0	1	1	0	1	1
0	0	1	0	1	1	0	1
0	0	1	1	1	0	0	1
0	1	0	0	0	1	0	1
0	1	0	1	1	0	1	1
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	1
1	0	0	0	0	1	0	1
1	0	0	1	0 X	0 X	1 X	1 X
1	0	1	0	0	0	1	0
1	0	1	1	1	0	0	0 1
1	1	0	0	0	1	0	1
1	1	0	1	1 X	0 X	1 X	1 X
1	1	1	0	0 X	0 X	1 X	0 X
1	1	1	1	1	0	0	1

13

Green $\rightarrow$ Yellow $\rightarrow$ Red $\rightarrow$ yellow $\rightarrow$ green

$\downarrow$
00

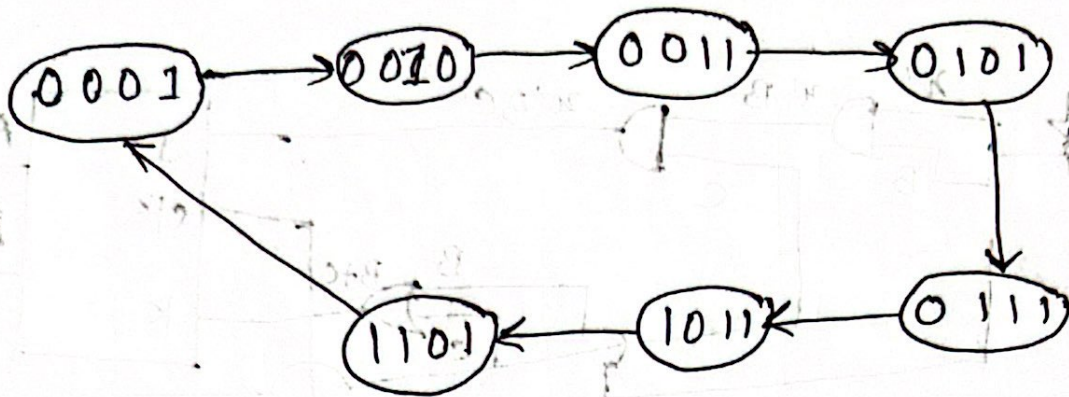
$\downarrow$
01

$\downarrow$
10



x	A	B	A+	B+	J _A	K _A	J _B	K _B
0	0	0	0	1	0	X	1	X
0	0	1	1	0	1	X	X	1
0	1	0	0	1	X	1	1	X
0	1	1	X	X	X	X	X	X
1	0	0	0	1	0	X	1	X
1	0	1	0	0	0	X	X	1
1	1	0	0	1	X	1	1	X
1	1	1	X	X	X	X	X	X

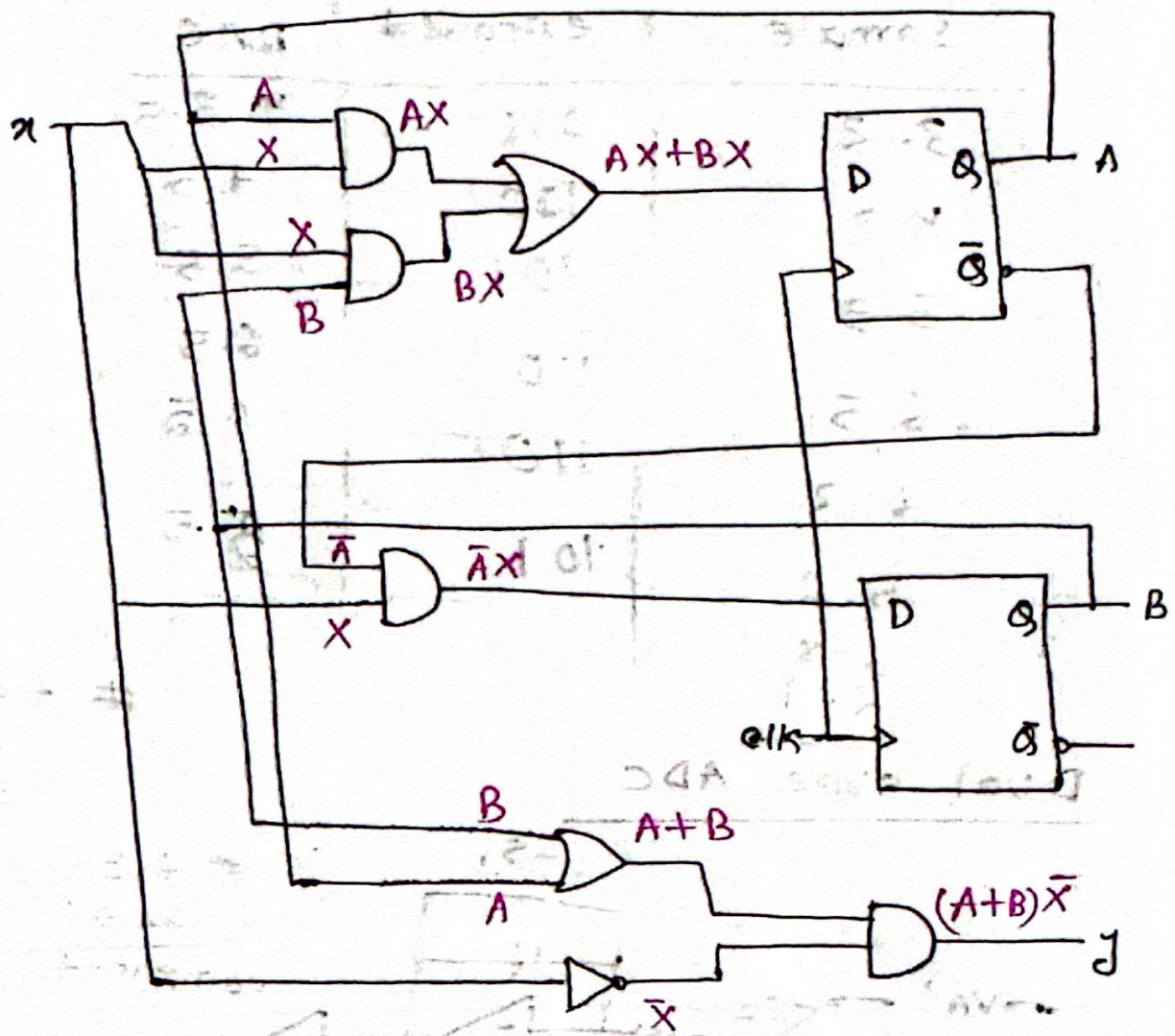
(19) $1 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 7 \rightarrow 11 \rightarrow 13 \rightarrow 1$



A	B	C	D	A ⁺	B ⁺	C ⁺	D ⁺	J _A	K _A	J _B	K _B	J _C	K _C	J _D	K _D
0	0	0	0	0	0	0	1	0	X	0	X	0	X	1	X
0	0	0	1	0	0	1	0	0	X	0	X	1	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	1	0	1	0	X	1	X	X	1	X	0
0	1	0	0	0	0	0	1	0	X	X	1	0	X	1	X
0	1	0	1	0	1	1	1	0	X	X	0	1	X	X	0
0	1	1	0	0	0	0	1	0	X	X	1	X	1	1	X
0	1	1	1	1	0	1	1	1	X	X	1	X	0	X	0
1	0	0	0	0	0	0	1	X	1	0	X	0	X	1	X
1	0	0	1	0	0	0	1	X	1	0	X	0	X	X	0
1	0	1	0	0	0	0	1	X	1	0	X	X	1	1	X
1	0	1	1	1	1	0	1	X	0	1	X	X	1	X	0
1	1	0	0	0	0	1	1	X	1	X	1	1	X	1	X
1	1	0	1	0	0	0	1	X	1	X	1	0	X	X	0
1	1	1	0	0	0	0	1	X	1	X	1	X	1	1	X
1	1	1	1	0	0	0	1	X	1	X	1	X	1	X	0

Assignment 4

⑥



i) 2 flip flops are there

$$D_a : Ax + Bx$$

$$D_b : \bar{A}x$$

$$Y : (A+B)\bar{X}$$

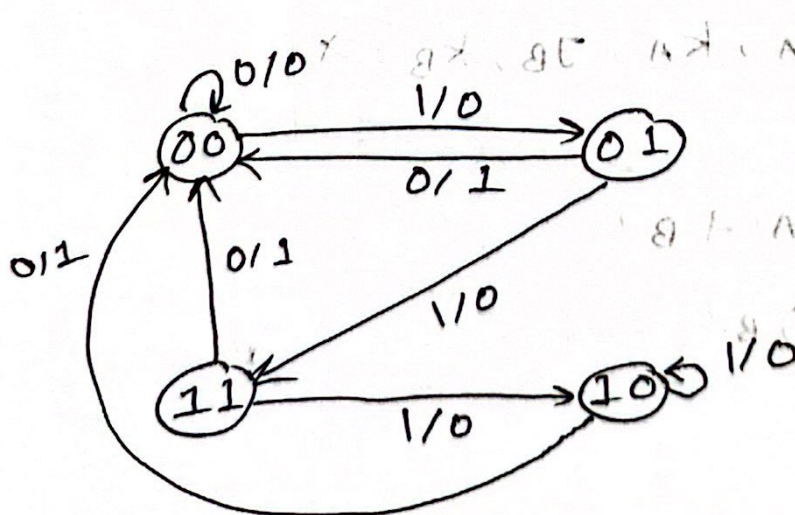
State table

A	B	x	Da	Db	y	A ⁺	B ⁺
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	0	0	0	1	0	0
0	1	1	1	1	0	1	1
1	0	0	0	0	1	0	0
1	0	1	1	0	0	1	0
1	1	0	0	0	1	0	0
1	1	1	1	0	0	1	0

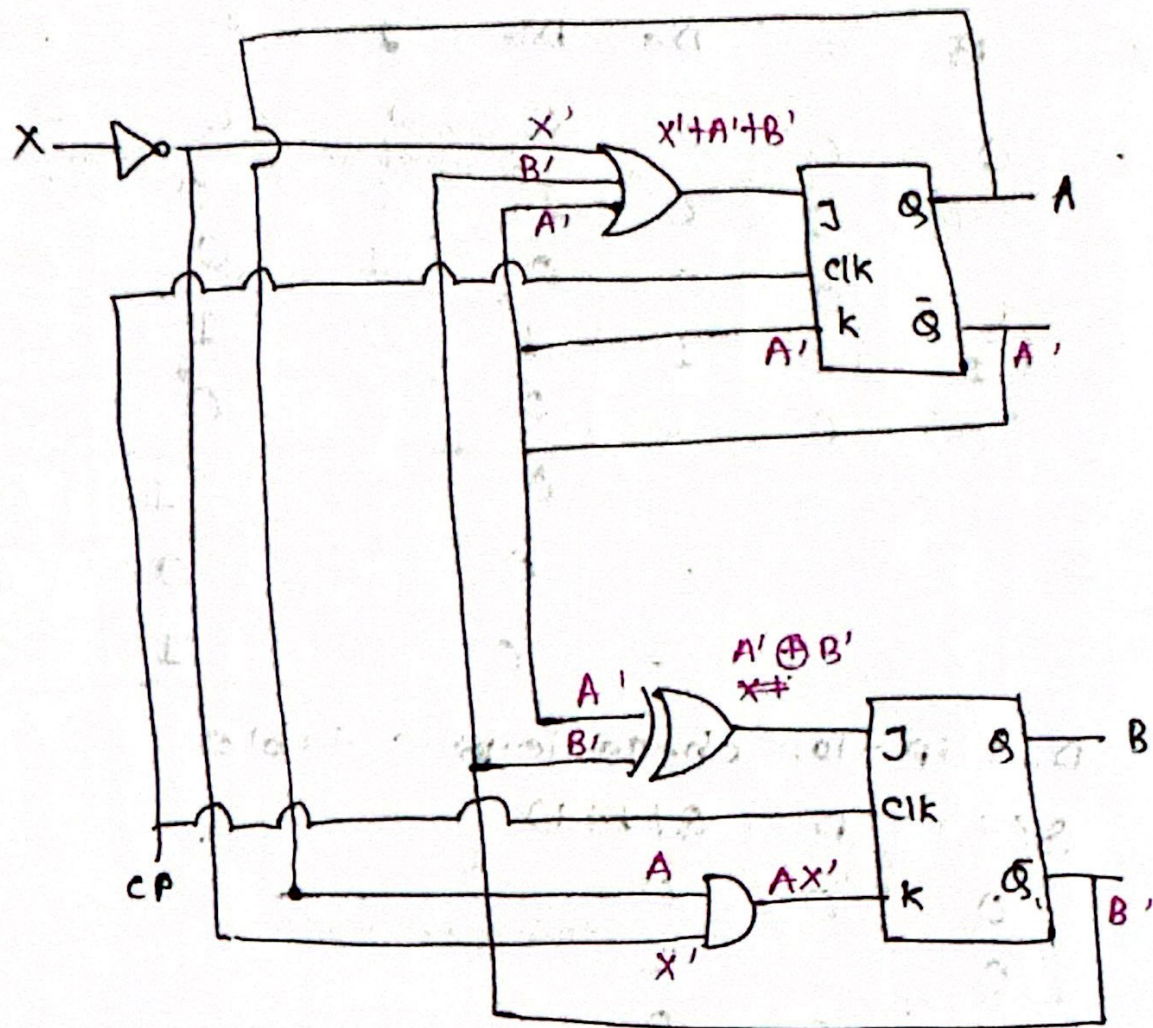
D flip-flop characteristic table

Q(t)	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

State diagram:



7



Present state : A , B

Next state : A + , B +

Input : J_A , K_A , J_B , K_B , X

Output :

$$J_A = X' + A' + B'$$

$$J_B = A' \oplus B'$$

$$K_A = A'$$

$$K_B = AX'$$

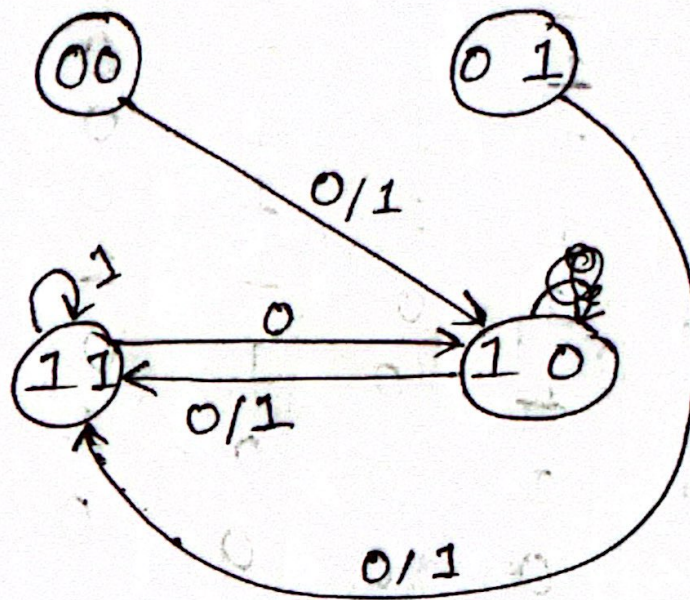
State table

A	B	x	J_A	K_A	J_B	K_B	A^+	B^+
0	0	0	1	1	0	0	1	0
0	0	1	1	1	0	0	1	0
0	1	0	1	1	1	0	1	1
0	1	1	1	1	1	0	1	1
1	0	0	1	0	1	1	1	1
1	0	1	1	0	1	0	1	1
1	1	0	1	0	0	1	1	0
1	1	1	0	0	0	0	1	1

Jk Characteristic table

$Q(t)$	J	K	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

State diagram :



Practice sheet (memory)

* How many address line do we need for a 64 MB RAM with 16 bit/word?

Capacity = no of word $\times$ bit per word

$$\Rightarrow 64 \text{ MB} = Y \times 16$$

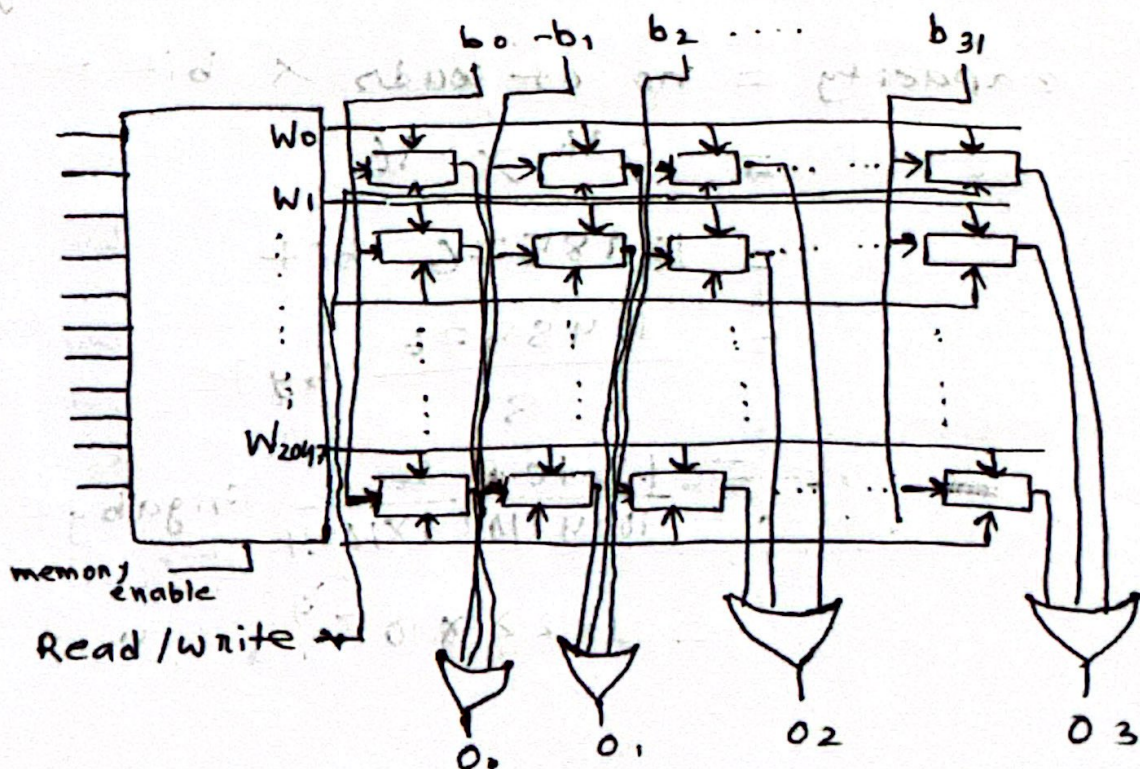
$$\Rightarrow 64 \times 8 \times 1024 \times 1024 = 16 Y$$

$$\Rightarrow Y = 33554432$$

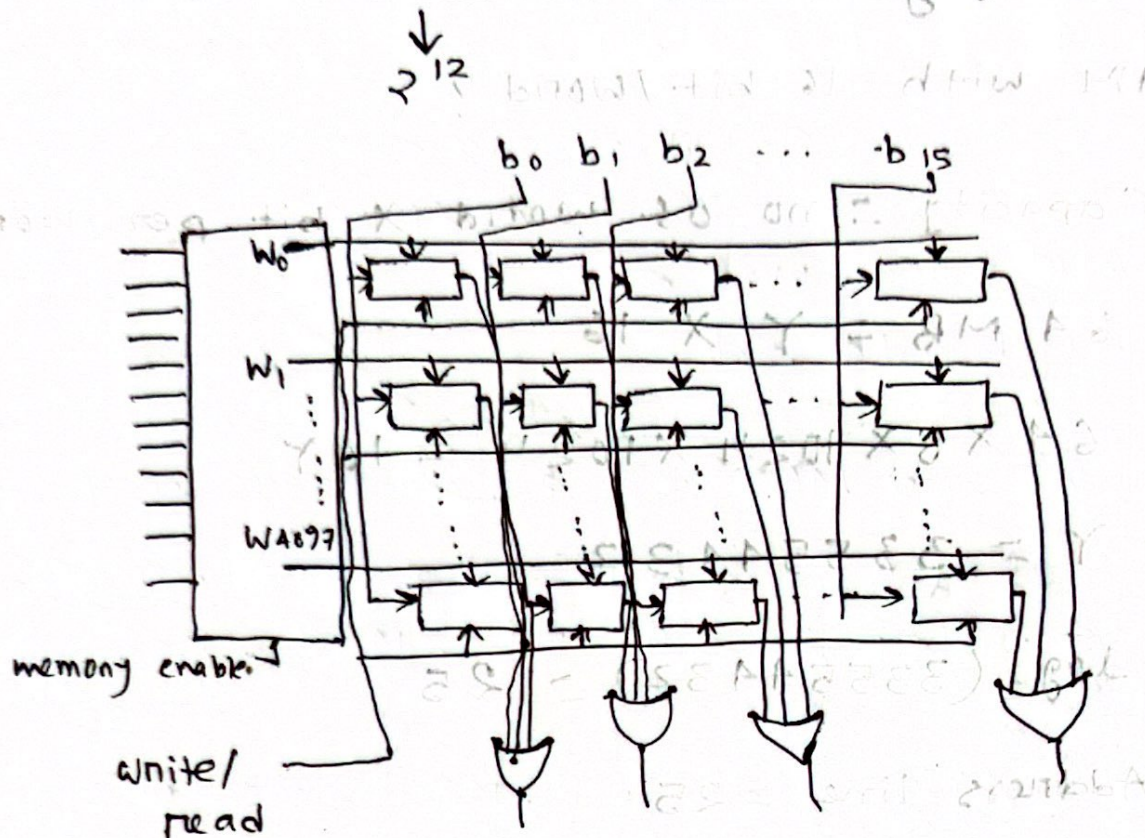
$$\log_2(33554432) = 25$$

Address line = 25

* Design a 2048 $\times$ 32 RAM showing internal details



* Design a 1024×16 RAM showing internal details



* What is the capacity for a $2^{16} \times 16$ RAM in Gigabytes?

capacity = no of words $\times$ bit per word

$$= 2^{16} \times 16$$

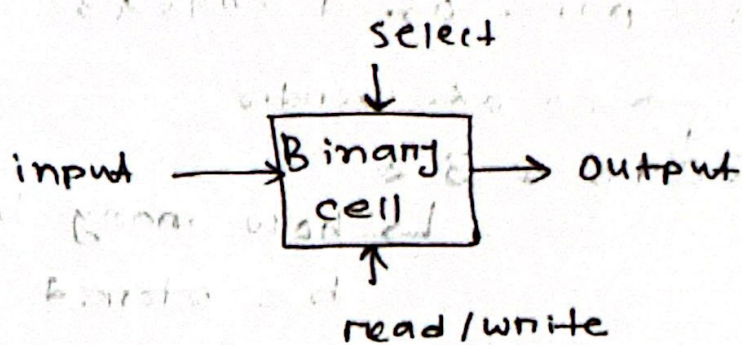
$$= 1048576 \text{ bit}$$

$$= \frac{1048576}{8} \text{ byte}$$

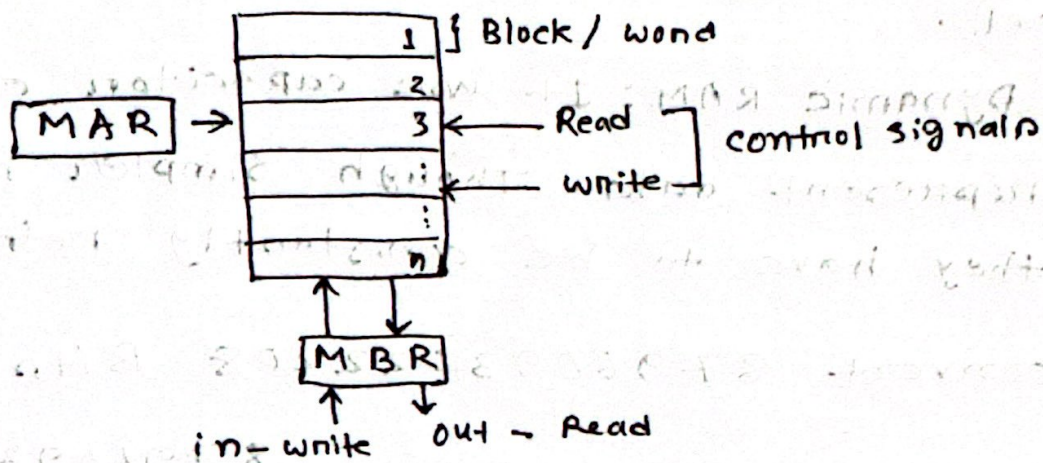
$$= \frac{131072}{1024 \times 1024 \times 1024} \text{ Gigabytes}$$

$$= 1.22 \times 10^{-4}$$

* Draw the block diagram of a binary cell.



* Draw and explain the functionalities of MAR and MBR.



* What type of information can you get from the '2048 x 32' part of a 2048 x 32 RAM?

→ no of words
2048 x 32
→ how many bits can be stored in a word

* How many types of RAM are there? explain

- There are two types of RAM.

i) Static RAM: It uses flip-flops as the memory cell.

ii) Dynamic RAM: It uses capacitors to represent data. Though simpler in circuitry, they have to be constantly refreshed.

* convert 8796093022208 Bits to Gigabyte

$$8796093022208 \text{ bits} = \frac{8796093022208}{8 \times 1024 \times 1024 \times 1024}$$

$$= 1024 \text{ Gigabytes}$$

* convert 3 Terabytes to bits.

$$3 \text{ Terabyte} = 3 \times 8 \times 1024 \times 1024 \times 1024 \times 1024$$

$$= 2.64 \times 10^{13} \text{ bits.}$$